When Yield Curves Invert Together: Currency and Carry Trade Crash Risk*

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Abstract

A currency is exposed to risk specific to its own country and to risk that is global. A single country's yield curve inversion can reflect either kind, but when several countries' curves invert at the same time, the likelier cause is a global shock. I build a monthly state variable from synchronized inversions across the non-dollar G10 currencies' yield curves and show that it marks periods of elevated currency crash risk. When the state is on, depreciation follows exposure. Currencies fall in the order of their interest rates and of their sensitivity to global equity markets, their currency betas, so the losses concentrate in the high-interest-rate currencies that a carry trade holds. The state raises the probability of a crash rather than the volatility of returns, survives standard global risk controls, and marks the months in which the forward premium puzzle reverses. Because the interest rate differential is the compensation for bearing exactly this risk, hedging it is a two-sided decision, and the shape of the inverted curves helps indicate whether the compensation is adequate. Inversions arriving with the whole curve sloping down from a high level accompany inflation normalizations, in which the differential is at its largest and the crash risk at its smallest, so a carry position that stays invested through them improves on one that exits whenever risk is elevated.

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1 Introduction

A currency is exposed to two kinds of risk, risk specific to its own country and risk that is global. The yield curve offers a country-by-country reading of the news behind both, the expected path of policy rates. Under the expectations hypothesis a curve inverts when markets expect rates to fall, and rates are expected to fall either because activity is expected to weaken or because a rate that has been raised well above neutral to bring inflation down is expected to normalize; the inverted curve is the classic recession predictor (Harvey, 1988; Estrella and Hardouvelis, 1991; Estrella and Mishkin, 1998). A single country's inversion, the U.S. curve's included, can therefore come from either source, a domestic credit cycle inverting one curve as readily as a global scare does. But when several countries' curves invert at the same time, the likelier cause is a shock that is common to all of them. This paper builds that observation into a monthly state variable, a cluster of inverted yield curves across the G10, which I call the IYC signal, for *inverted yield curves*, and shows that it marks in advance the months in which currencies crash.

The portfolio for which such a state matters most is the currency carry trade, which borrows in low-interest-rate currencies to invest in high-interest-rate ones, earns the interest differential on average, and gives some of it back in crashes. A diversified carry portfolio already handles the country-specific risk, since three currencies per leg, re-sorted every month, absorb idiosyncratic shocks. What it cannot absorb is a common shock that hits its whole long leg at once, so its risk-management problem is one question, asked repeatedly—is the exchange-rate leg about to move by more than the differential covers? The right conditioning statistic is one of the common component, and the count of simultaneous inversions among the nine non-dollar G10 curves is that statistic; two is the smallest count at which a systemic explanation is more likely than a coincidence. The IYC signal adds a trader's discipline to the count. A curve's inversion is counted from the month after it is first observed until that curve's slope has steepened for two consecutive months, the conventional confirmation of a turning point, and the signal is on while at least two inversions are counted. It therefore switches on when a fresh inversion completes a cluster and off only once re-steepening has been confirmed curve by curve. Figure 1 shows the signal against the carry trade's cumulative returns. The losses sit inside its episodes, and the five worst months of the sample all fall inside a state that occupies a fifth of the calendar.

The exit rule matters more than it looks. Delivered easing re-steepens curves mechanically, so a plain inversion count switches off precisely at the crash apex. The three worst carry months in the sample—October 2008, March 1995, March 2020—occur while the IYC signal is on but after the naive count has already fallen below two. A rule built for a trader's re-entry problem turns out to solve a classifier's measurement problem, and the distinction between the disciplined signal and the naive count runs through the paper. The signal's content does not rest on those three months, or on any crash month. Dropping each portfolio's bottom-decile months from both states, the carry portfolio's signal-month spot deficit is still -7.2%/yr (wild $p = 0.006$; Table 4)—the state shifts the whole return distribution, not just the tail the latch is built to catch.

The empirical content is straightforward to state. Inside the signal's 92 months (of 458), every conditional moment of the carry trade deteriorates at once. The total excess return swings from $+6.5$ to -4.9 percent per year, skewness and expected shortfall worsen sharply, and the exchange-rate leg loses about 13 percent per year relative to no-signal months. That loss strengthens rather than weakens when volatility and financial-conditions gauges are held fixed, because the state has no second-moment signature for those gauges to span—it shifts the mean and raises the probability of a crash month with no significant change in variance. In the language of the disaster-risk literature (Rietz, 1988; Barro, 2006; Wachter, 2013), the signal isolates a time-varying disaster probability that is observable in public yield data rather than filtered from consumption or extracted from option prices. The oldest expression of the fact is the regression of exchange-rate changes on the lagged interest differential, whose slope should be -1 under uncovered interest parity (UIP) but has sat near zero since Fama

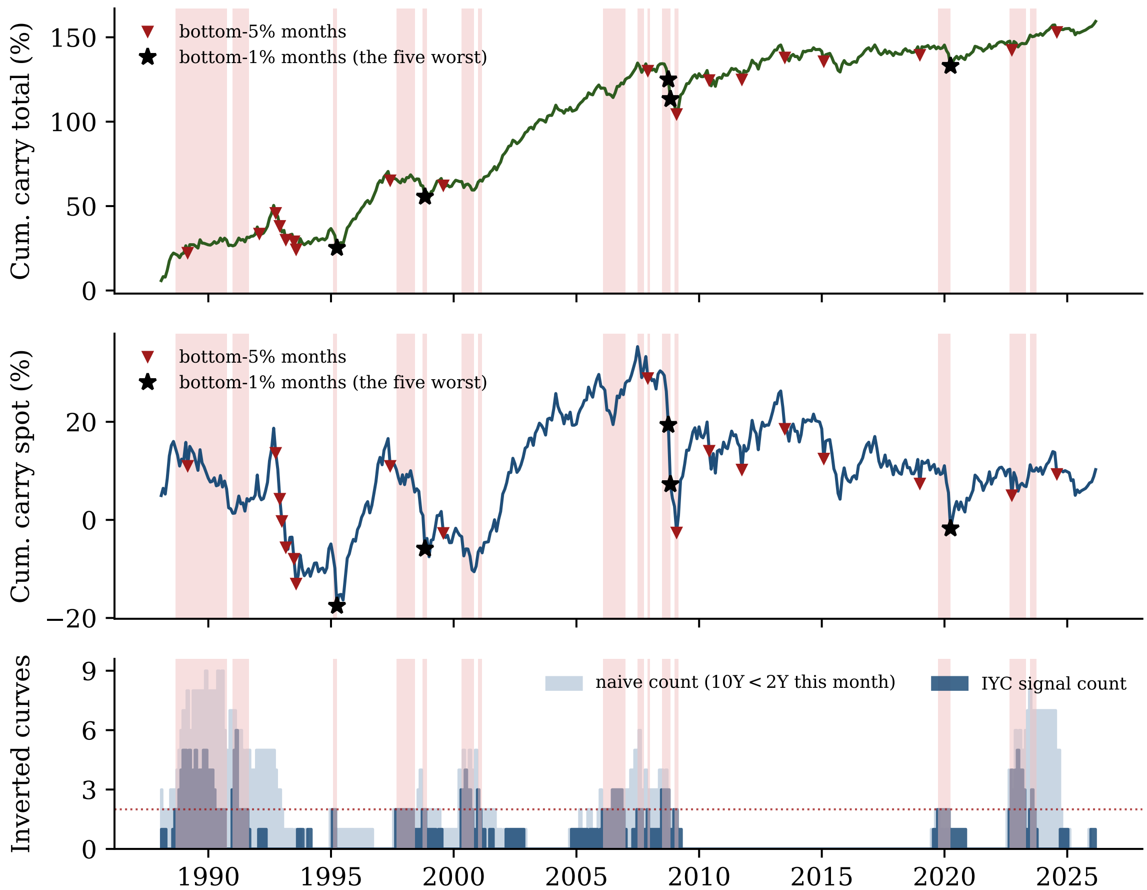

Figure 1: The carry trade and the IYC signal. Top to bottom: the carry portfolio's cumulative total return, its cumulative spot return, and two counts of inverted G10 curves (the nine non-dollar members)—the *naive* count (light), curves with the 10-year yield below the 2-year this month, and the *IYC signal* count (dark), the inversions the signal counts, entered fresh and not yet confirmed re-steepened—with the cluster threshold of two dotted. IYC signal episodes are shaded throughout; triangles mark each return series' own bottom-5% months, the crash months (the two sets share 22 of their 23), and black stars its bottom-1% months, the five worst of the sample, all of which fall inside the signal's lagged state (three of them—March 1995, October 2008, and March 2020—are the first month after a shaded band, when the lagged state is still on). The shading is, by construction, the months in which the signal count is at least two, and the gap between the two bars is the signal's discipline in both directions. Standing naive inversions with no *fresh* entry are not counted (1991–92, 2023–24), and counted inversions persist through the delivery phase after the naive count has receded (early 1995, late 2008, early 2020)—the latch at work. Shading is the information-time state; every regression uses it lagged one month.

(1984), the forward premium puzzle. Outside the signal the puzzle holds in its classic form, with a Fama slope of about +0.5; inside it the slope is about −1.3, so that UIP is not merely restored but overshot. The signal, in other words, identifies the months in which the puzzle mean-reverts, the payback states that the peso-problem reading of Burnside et al. (2011) requires and that the unconditional regression averages away.

The loss lands where the exposure is. Persistently high-rate economies are the commodity exporters and small open economies that are long the global cycle (Ready et al., 2017; Della Corte et al., 2016), and several complementary accounts—country size (Hassan, 2013), position in the trade network (Richmond, 2019), the static ranking of Hassan and Mano (2019)—load the same currencies on world risk. Their equity betas confirm it. The signal's currency-level loss scales with beta and vanishes at zero beta, a high-beta-sorted portfolio loses on the signal with the same profile as the carry portfolio, and measured activity deteriorates after the signal fires, consistent with the business-cycle exposure of currency premia documented by Colacito et al. (2020). The identification is about the cluster rather than about inversions as such. A month with exactly one counted inversion is a good carry month, because a lone inversion is typically a high-rate country's own cycle, which the portfolio collects rather than suffers, and the sign flips at the second curve. In a horse race against the continuous global average slope, the U.S. curve, and all nine countries' own signals, the cluster's coefficient is unchanged or larger. The U.S. curve alone, the common practitioner proxy for a global signal, predicts nothing, because it misses two-thirds of the cluster's months, including the Asian crisis, the 1995 and 2020 episodes, and the fifteen months leading into September 2008. The rule has no fitted parameters, only the three design choices (two curves, two steepening months, the 10-year/2-year tenor) whose rationales Section 2 gives, and the two halves of the sample, before and after 2005, each show the loss, with the later half, which contains 2008 and 2020, the stronger of the two.

A trader who sees the cluster form can buy options against the tail (Burnside et al., 2011; Jurek, 2014; Farhi et al., 2015; Caballero and Doyle, 2012), unwind, or reverse into anti-carry. This paper studies the two hedges that need no option market, and its second theme is that the hedging decision is two-sided. The usual risk-management question is how to protect against the crash. But an unwind forgoes the differential, which is the compensation for bearing exactly this risk, so an unwind in a state where the differential would have covered the loss is a loss of its own. I call this carry-premium risk, the mirror image of crash risk; it grows with the length of the unwind, and an anti-carry position, which pays the differential and loses the rally, bears it twice on a false alarm. The investor's problem is therefore not simply whether to hedge when risk is high but whether the compensation is sufficient to justify remaining invested, and the carry trade is unusually suited to that question because its compensation, the differential, is observable in advance while the crash is not. The curve speaks to both sides. The count of inversions measures the risk side, when disaster risk is elevated; the differential and the *shape* of the average inverted curve inform the compensation side, whether the investor is paid to bear it. Two disciplines follow. On duration, the signal's exit rule restores the position for the recovery rather than leaving it stranded in cash, and the resulting exit strategy raises the carry trade's Sharpe ratio from 0.53 to 0.79 and halves the worst month. On state, when the whole curve slopes downward from a high level, the configuration of an inflation-fighting peak that is expected to normalize, the differential is at its largest and the exchange-rate loss at its smallest, and a carry overwrite that stays invested through such inversions raises the Sharpe ratio further, to 0.82 for the exit and to 0.90 for the anti-carry version, the larger gain for the position whose false alarms cost double; in the other inversions, the growth scares, the differential covers only a fraction of the loss. The downcurve regime rests on the two inflation eras the sample contains, and the compensation principle is more general than that particular implementation. The same principle has a comparative static. The seven floating emerging-market currencies enter nowhere in the signal's construction, which makes them an out-of-construction test, and all seven load negatively on it; in a portfolio that adds them to the G10 the differential is large in every state, so exiting on the signal leaves the mean essentially unchanged and trims the tail, anti-carry fails because it is short a very large differential, and the shape has no compensation regime left to distinguish.

Related literature

The paper's outcome variable belongs to the carry-trade literature. Uncovered interest parity predicts that high-interest-rate currencies depreciate by the interest differential; since Hansen and Hodrick (1980) and Fama (1984) the regression of exchange-rate changes on the lagged differential has returned a slope near zero rather than the minus one parity implies, the forward premium puzzle, so the carry trade earns the differential and, on average, a little more (Engel, 2016). The modern literature reads the puzzle as a risk premium and organizes currency returns into priced factors (Lustig and Verdelhan, 2007; Lustig et al., 2011; Menkhoff et al., 2012; Lettau et al., 2014; Dobrynskaya, 2014; Verdelhan, 2018). What gives the premium its bite is the shape of the payoff. Brunnermeier et al. (2008) show that carry returns are negatively skewed and crash-prone, with the crashes coinciding with unwinds of crowded positions and funding-liquidity spirals; the pattern recurs across a century of monetary regimes (Accominotti et al., 2019), and in the disaster models of Farhi and Gabaix (2016) and Chernov et al. (2018) the crashes are the realizations that the differential compensates for in advance.

That the yield curve forecasts activity is established for single countries (Harvey, 1988; Estrella and Hardouvelis, 1991; Estrella and Mishkin, 1998; Ang et al., 2006; Wright, 2006; Bauer and Mertens, 2018), with Campbell and Shiller (1991) supplying the expectations-hypothesis accounting and Hamilton and Kim (2002) and BIS (2019) the decomposition of the slope into expectations and term-premium components. The fact travels across countries (Plosser and Rouwenhorst, 1994; Estrella and Mishkin, 1997; Bernard and Gerlach, 1998; Chinn and Kucko, 2015), an average of term structures forecasts world growth (Harvey, 1991), and global factor models treat the world's curves as one object (Diebold et al., 2008). What that literature aggregates is predictive content country by country or a common level factor. The cross-sectional configuration—how many curves stand inverted at once—is not used as a state variable in it, for output or for currencies, and the finding that the U.S. curve alone carries no information for currency crashes is, to my knowledge, new. On the exchange-rate side, Backus et al. (2001) state the joint restriction between term-structure models and the forward premium, Bekaert et al. (2007) test UIP jointly with the term structure, Chernov and Creal (2023) price international curves and currencies together, and Ang and Chen (2010) show that curve levels and slopes predict currency returns country by country. That curve shape carries currency information is established bilaterally by Chen and Tsang (2013), who predict exchange rates with relative Nelson–Siegel factors, Dreher et al. (2020), who sort currencies on relative curvature, Baek et al. (2020), who relate butterfly gaps to carry premia, and Lloyd and Marin (2020), who price transitory components of relative curves; Lustig et al. (2019) show that carry premia shrink with bond maturity. On the macro side, curvature has policy-stance content (Mönch, 2012), and the common component of international curves is largely a world-inflation object (Jotikasthira et al., 2015), which is the premise that licenses reading the average G10 curve's shape as an inflation-regime indicator. The systemic component the cluster isolates is the bond-market counterpart of the common factor that the global-financial-cycle literature identifies from asset prices and capital flows (Rey, 2013; Miranda-Agrippino and Rey, 2020) and that intermediary-based pricing traces to dealer balance sheets (He et al., 2017; Gabaix and Maggiori, 2015); the cluster reads it off expected policy paths, in advance.

Relative to the literature that conditions carry on observable states, the state here is built from bond markets' expected policy paths rather than from realized risk gauges, rate levels, or portfolio composition. Clarida et al. (2009) and Christiansen et al. (2011) show that carry's returns and loadings are regime-dependent with volatility as the regime variable, Berge et al. (2011) evaluate a menu of timing signals, Cenedese et al. (2014) and Bakshi and Panayotov (2013) predict carry with FX-risk, commodity, and liquidity variables, Lustig et al. (2014) time the dollar carry trade on the average forward discount, Barroso and Santa-Clara (2015) build optimal portfolios from such characteristics, Bekaert and Panayotov (2020) show that which currencies are held governs crash exposure, and Jordà and Taylor (2012) show that fundamentals-adjusted strategies avoid much of the tail. None of these prices the cost of the unwind, which is what the two-sided discipline here is built around; the practitioner literature documents that cost mainly as the magnitude and duration of carry unwinds (Melvin and Shand, 2017), and Daniel et al. (2017) measure the drawdowns. Three papers come closest. Lustig et al. (2014) condition on a cross-country average of rate information, but on its level rather than the curve's shape, and for the dollar rather than high-minus-low carry. Andrews et al. (2024) show that a carry-family strategy's compensation depends on a global inflation-expectations state, for bond slope carry and with the regime read from macro expectations around a pre- and post-2008 break rather than from an observable curve configuration in real time. And Bakshi and Panayotov (2013) provide the closest precedent for the overwrite's form, an on/off overlay built on commodity, volatility, and liquidity predictors rather than any yield-curve object. Kaebi and Martins (2025) decompose the differential into trend and cycle and find that the persistent component owns carry's unconditional profitability with less negative unconditional skewness; the conditional results here complete that picture, since a state-contingent mean drift is invisible to unconditional skewness. Risk-based accounts of the differential more broadly run through habits and long-run risks (Verdelhan, 2010; Bansal and Shaliastovich, 2013), and Koijen et al. (2018) document carry's generality across asset classes. The inflation-fighting-versus-growth-scare typing of inversions exists in market commentary; its formalization and pricing appear to be new.

The rest of the paper proceeds as follows. Section 2 presents the framework and the theory in one arc, from the portfolio construction and the signal's logic to the pricing benchmark and its testable implications; Section 3 describes the data and the signal; Section 4 documents the mechanism, who loses and why; Section 5 reports the carry trade's conditional moments, state regressions, and distribution in and out of the signal and prices the hedging strategies and the carry overwrite; Section 6 identifies the cluster against a country's own curve and the U.S. curve; Sections 7–9 take the signal to emerging markets, to the portfolio that includes them, and to the stock market; Section 10 confronts the signal with the standard risk gauges, and Section 11 collects the robustness evidence.

2 Framework and Theory

This section develops the paper's economic logic once, from the carry trade's accounting to the pricing benchmark, so that the empirical sections can be read against it. The reduced-form statements are accounting identities and simple statistics; the formal results are propositions in the no-arbitrage kernel setting of the currency literature (Lustig and Verdelhan, 2007; Lustig et al., 2011), which assumes nothing about preferences beyond a positive kernel that is high in the recession state (risk neutrality enters only in the trader's decision rule of Proposition 4). Each formal result is tagged as *derived* or *calibrated*; proofs, derivations, and every calibrated piece are in Appendix B.

2.1 Carry crashes and common exposure

Portfolios follow the standard currency-portfolio setting of Lustig et al. (2011), from the U.S. investor's side. Let $s_{i,t}$ be the log spot price of currency i in dollars and $i_{i,t}$ its one-month money-market rate. The monthly excess return on a long position in currency i funded in dollars is

$$rx_{i,t+1} = \underbrace{\tfrac{1}{12}\left(i_{i,t} - i_{US,t}\right)}_{\text{income (carry) leg}} + \underbrace{\Delta s_{i,t+1}}_{\text{spot leg}}, \tag{1}$$

which under covered interest parity equals the forward discount less the realized depreciation, $f_{i,t} - s_{i,t+1}$; the income leg is known at t and the spot leg is the exchange-rate realization. (Post-2008 deviations between deposit rates and forward-implied rates are documented by Du et al. (2018); the income leg here is the money-market differential throughout.) The *carry trade* is the interest-rate-sorted long–short portfolio of Lustig et al. (2011), their high-minus-low currency factor. Each month the currencies are ranked on $i_{i,t}$, and the portfolio goes long the k highest-rate currencies and short the k lowest-rate ones, equally weighted within each leg, so that for a $k \times k$ construction

$$HML^{carry}_{t+1} \;=\; \frac{1}{k}\sum_{i\in\,\text{top-}k(i_{i,t})} rx_{i,t+1} \;-\; \frac{1}{k}\sum_{i\in\,\text{bottom-}k(i_{i,t})} rx_{i,t+1} \;=\; C_t + X_{t+1}, \tag{2}$$

where the total return splits, by applying the same weights to the two components of (1), into the income leg $C_t = \frac{1}{12k}\big(\sum_{\text{top-}k} i_{i,t} - \sum_{\text{bottom-}k} i_{i,t}\big)$ and the spot leg $X_{t+1} = \frac{1}{k}\big(\sum_{\text{top-}k} \Delta s_{i,t+1} - \sum_{\text{bottom-}k} \Delta s_{i,t+1}\big)$. The paper uses $k = 3$ throughout, for the nine G10 currencies and for the sixteen-currency portfolio of Section 8 alike; with nine currencies the 3×3 sort is the tercile split of the standard setting. The *high-beta portfolio* replaces the sorting variable with the currency's *currency beta*—its exposure to the equity market, in the sense of Lettau et al. (2014) and Kremens and Martin (2019) (the currency-beta discussion below prices it). For each currency, $\hat{\beta}_{i,t}$ is the slope from the expanding-window regression

$$rx_{i,\tau+1} \;=\; a_i + \beta_i\, r^M_{\tau+1} + \varepsilon_{i,\tau+1}, \qquad \tau \le t-1, \text{ minimum 36 months}, \tag{3}$$

where r^M is the U.S. equity market return—equivalently, $\hat{\beta}_{i,t}$ is the expanding-sample covariance of the currency's return with the market divided by the market's variance, using only data through $t-1$—and HML^{β}_{t+1} sorts on $\hat{\beta}_{i,t}$ exactly as in (2). The expanding monthly window is chosen for real-time coverage from 1988 and for stability of the sort, so that the beta captures the slow-moving growth-exposure trait that the mechanism section measures rather than month-to-month variation; Appendix G repeats the headline cells under an MSCI World benchmark (daily rolling-window betas are reported in the replication package). Both portfolios are observable at the start of the return month. The two sorts are close cousins, which is the mechanism section's point—the rate ranking and the beta ranking are two measurements of the same growth-exposure trait.

Write $r^c_{t+1} = C_t + X_{t+1}$ for the carry trade's total return in (2). The income leg C_t is known at t and non-negative by construction of the sort, so the trade loses in a month if and only if

$$X_{t+1} \; < \; -C_t, \tag{4}$$

and every carry crash is an exchange-rate event that overwhelms the differential. In the sample the income leg averages +3.9%/yr and is positive in every month (its minimum is +0.7%/yr), while the spot leg carries all of the strategy's tail (Section 5). The risk-management question is therefore not whether currencies move but whether the differential covers the move.

The carry portfolio already handles idiosyncratic risk on its own, which is why the signal is a cluster rather than a country's own inversion. Write the spot return of currency i as the sum of a common and an idiosyncratic component,

$$\Delta s_{i,t+1} \; = \; -\,\beta_i\, g_{t+1} \; + \; e_{i,t+1}, \qquad \beta_i > 0, \tag{5}$$

where g_{t+1} is a realized global growth disappointment and β_i currency i's exposure to it. The equally weighted long–short portfolio of (2) then has spot return and variance

$$X_{t+1} \; = \; -\big(\bar\beta^L - \bar\beta^S\big)\, g_{t+1} \; + \; \big(\bar e^L_{t+1} - \bar e^S_{t+1}\big), \qquad \mathrm{Var}(X) \; = \; \big(\bar\beta^L - \bar\beta^S\big)^2 \sigma_g^2 \; + \; \tfrac{2}{k}\,\sigma_e^2, \tag{6}$$

for k currencies per leg with i.i.d. idiosyncratic shocks. The second term is what diversification and the monthly re-sort take care of; it shrinks with k, has zero mean, and no ex-ante signal can be about it. The first term is the only component with a systematic loss, and the sort loads on it by construction because the rate ranking tracks the exposure ranking (rank correlation 0.72, Section 4). A risk-management signal must therefore be a statistic of the *common* component of expected growth, which under the expectations hypothesis of Section 2.2 is what the cross-country configuration of slopes measures.

Who loses. Currencies load differently on global growth. Commodity exporters and small open economies with persistently high interest rates are long the global cycle; funding havens with persistently low rates are short it. When a slowdown is delivered, exchange rates reprice in proportion to that exposure—(5) with the β_i ranked as the differentials are ranked—so the crash concentrates in the carry trade's long leg, because the trait that generates the differential generates the exposure. Section 4 documents the chain, showing that the rate ranking coincides with the measured growth-exposure ranking, that the signal's loss scales with that exposure and vanishes without it, and that measured activity deteriorates after the signal fires.

The currency beta. The exposure β_i of (5) has an asset-pricing counterpart. Under a one-factor pricing kernel whose factor is the global equity market, the expected excess return on currency i is its covariance price,

$$\mathbb{E}_t\big[rx_{i,t+1}\big] \;=\; \beta_i^M\,\lambda_t^M, \qquad \text{so that} \qquad \mathbb{E}_t\big[\Delta s_{i,t+1}\big] \;=\; \beta_i^M\,\lambda_t^M - C_{i,t}, \tag{7}$$

where β_i^M is the currency's beta on the market return, λ_t^M the market risk premium, and $C_{i,t}$ the currency's interest rate differential against the dollar. Kremens and Martin (2019) make the covariance observable. Their quanto theory expresses the expected appreciation of a currency as the interest differential plus the risk-neutral covariance of the currency with the U.S. stock market, recovered from quanto index contracts, and they find that the covariance term forecasts exchange rates with the sign (7) implies. Two things follow for the paper. First, the differential and the beta are two measurements of one trait. A currency whose return covaries with global risk pays a high differential in the survival state and loses in the state in which the risk is realized, so the rate ranking and the beta ranking should coincide, which Section 4 confirms (rank correlation 0.72). Second, the signal's loss should scale with β_i^M and vanish at $\beta_i^M = 0$, because a growth scare is a realization of the factor (5) and a currency without exposure has nothing to reprice. The high-beta portfolio of (3) sorts on an expanding-window estimate of β_i^M, and its role in the paper is to measure the same exposure the carry sort measures, from asset prices rather than money markets; the conditional Fama slopes of Section 4 are the regression image of the same relation, since a differential that compensates a covariance with global risk predicts depreciation precisely when the covariance is realized.

2.2 Synchronized yield curves as a signal of the common state

Let $y_{i,t}^{(n)}$ be country i's n-period yield. Under the expectations hypothesis with a slow-moving term premium τ_i,

$$y_{i,t}^{(10Y)} - y_{i,t}^{(2Y)} = \underbrace{\frac{1}{H}\sum_h \mathbb{E}_t[r_{i,t+h}] - r_{i,t}}_{\text{expected policy path}} + \tau_i + \varepsilon_{i,t}. \tag{8}$$

The slope is negative when the market expects the policy path to fall below its current level. Central banks ease when they expect activity to weaken or when policy has been set well above neutral, so an inversion is a market-implied forecast of easing—in the first case the recession-signal logic of Harvey (1988) and Estrella and Mishkin (1998), in the second a normalization. This is the only asset-pricing input the signal needs; the term premium need not be zero, only slow relative to the expected-path component. The two causes, a *growth scare* and *inflation fighting*, have opposite implications for a carry position, and Section 2.5 shows that the shape of the curve separates them.

The count as the statistic. The count of inversions is the statistic of the common component that the portfolio algebra of Section 2.1 calls for. Write the expected-slowdown component of country i's slope as $\lambda_i g_t^e + \epsilon_{i,t}$, with g_t^e the common expected-growth shock and $\epsilon_{i,t}$ idiosyncratic. A single inversion can come from either source, but with independent $\epsilon_{i,t}$ the probability that idiosyncratic shocks alone invert two curves at once is an order smaller than for one, while a common shock moves all curves together. The second simultaneous inversion is therefore the informative one, and two is the smallest count that is not one country's news. Months with three or more live inversions—*live* in the sense of the rule defined in the next paragraph—sit deeper into an episode—their median episode age is six months, against three for the exactly-two months, and eleven of the fifteen episodes begin at exactly two—by which point much of the damage documented in Section 4 has occurred, and within the signal a three-or-more dummy adds nothing (wild $p \geq 0.27$, with or without episode age held fixed; computations in the replication package). What matters after the second curve is the cluster's freshness, not its width. The data sharpen the threshold itself. A month with exactly one live inversion turns out to be a *good* carry month and the sign flips at the second curve (Section 6), so accepting a single inversion averages two opposite states, while requiring three or four arrives too late and leaves too few months (Appendix C). Conditional on the cluster, the identities of the inverted countries—including the United States—should carry little extra information, a prediction Section 6 tests directly.

The IYC signal: fresh entry, confirmed exit. The signal is built curve by curve from 10-year/2-year inversions. Let $s_{i,t} = y_{i,t}^{10Y} - y_{i,t}^{2Y}$ be curve i's slope at the end of month t and $I_{i,t} = \mathbf{1}\{s_{i,t} < 0\}$ its inversion. Each curve carries a *live* indicator $L_{i,t}$ and a counter $k_{i,t}$ of consecutive steepening months, $k_{i,t} = (k_{i,t-1} + 1)\,\mathbf{1}\{\Delta s_{i,t} > 0\}$, and

$$
\begin{aligned}
L_{i,t} &= \begin{cases} 1 & \text{if } L_{i,t-1} = 0 \text{ and } I_{i,t-1} = 1,\ I_{i,t-2} = 0 \quad \text{(fresh inversion)}, \\ 0 & \text{if } L_{i,t-1} = 1 \text{ and } k_{i,t} \geq 2 \quad \text{(released)}, \\ L_{i,t-1} & \text{otherwise}, \end{cases} \\
N_t &= \sum_{i=1}^{9} L_{i,t}, \qquad S_t = \mathbf{1}\{N_t \geq 2\}.
\end{aligned}
\tag{9}
$$

A curve's inversion becomes live the month after it is first observed, and the signal is on while at least two inversions are live. A live inversion is released only after two consecutive months in which that curve's slope steepens, whether or not the slope is back above zero, and a released curve that stays inverted cannot re-enter until it has un-inverted and inverts afresh. The signal S_t uses data through month t and positions the portfolio for month $t + 1$. The same recursion applied to one curve defines that country's *own signal*, the rival of Section 6.

Both halves of the rule are the trader's discipline. The entry is fresh because an old cluster's information is already in prices; what matters is the arrival of a new systemic scare, not the continuation of an old one. The exit is a confirmation. Two consecutive steepening months mark the turning point of a curve's expected-rate path, the point at which the easing that constitutes the crash is being delivered. The confirmation is kept as short as the problem's asymmetry allows, because re-entering into a still-falling path costs a crash while waiting one more month costs one month of differential (Section 2.5). One month of re-steepening is frequent noise inside episodes, three sits out too much of the recovery, and two is the same convention that defines a recession by two consecutive quarters; Appendix C tabulates the alternatives. The confirmation—the *latch* that holds a curve live until its release is confirmed—has a second virtue. Delivered easing re-steepens curves mechanically, so any measure that requires curves to remain inverted switches off during the delivery phase, precisely when the losses arrive; the latch stays on through it. The naive alternative—a curve counts while its slope is below zero, the signal is on while at least two curves count—is the benchmark of Appendix D, and Figure 2 shows on one episode what the rule buys.

Two primitives give the count-as-statistic argument above its formal version, with the global state as the object the count estimates.

Assumption 1 (Latent cycle)**.** *A global state $g_t \in \{N, R\}$ (normal, recession) follows a two-state Markov chain with rare onset rate $\Pr(R \mid N) = p_{NR} \ll 1$ and small stationary prior $q_0 = p_{NR}/(p_{NR} + p_{RN})$.*

Assumption 2 (Slopes)**.** *Country i's 10-year/2-year slope is $S_{i,t} = \mu_{g_t} + \varepsilon_{i,t}$ with $\mu_R = \mu_N - \theta$, $\theta > 0$; the idiosyncratic $\varepsilon_{i,t}$ is i.i.d. normal across the $n = 9$ countries. Inversion is $I_{i,t} = \mathbf{1}\{S_{i,t} < 0\}$, with per-country inversion probabilities $q_R > q_N$ and count $k_t = \sum_i I_{i,t}$.*

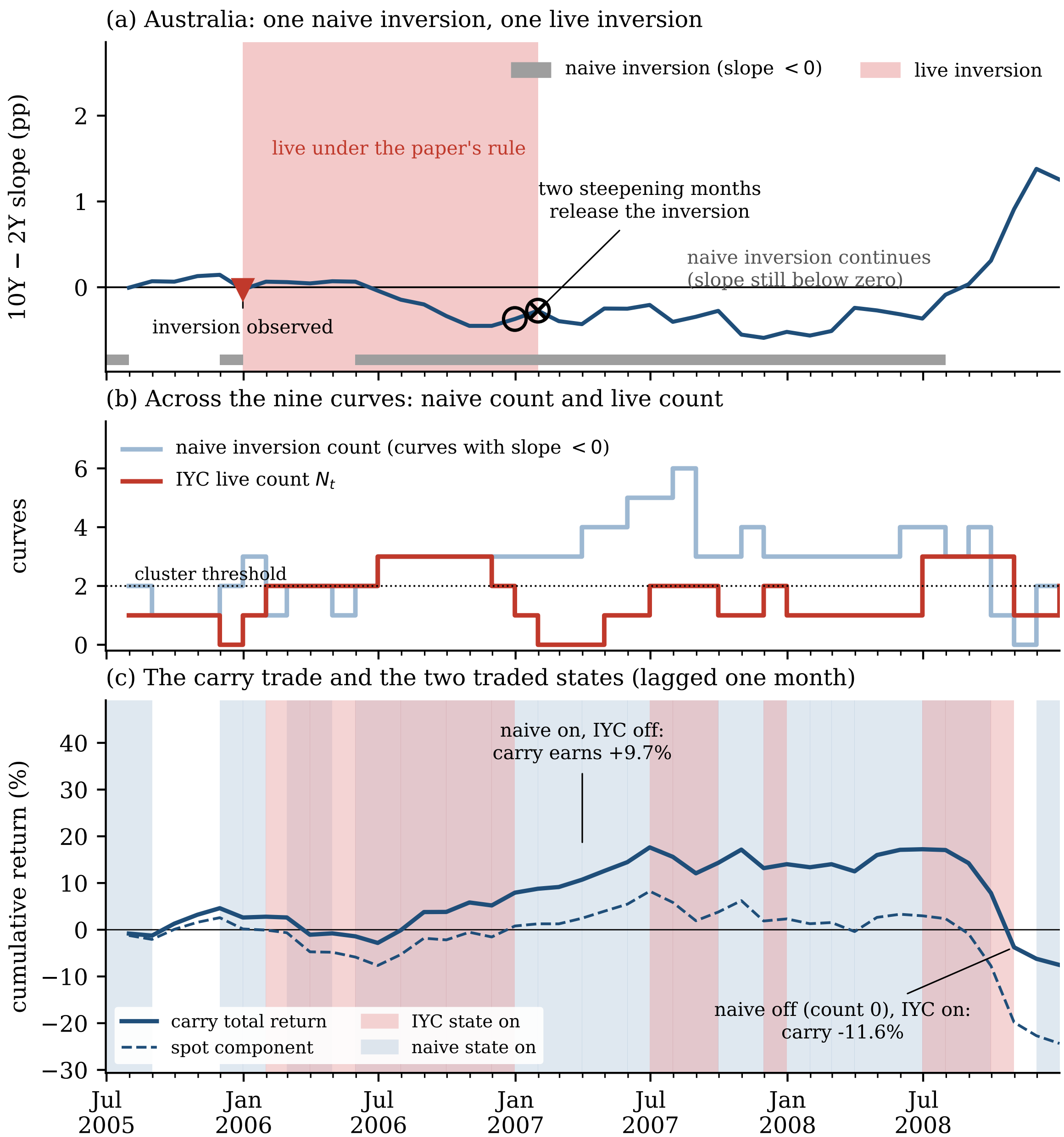

Figure 2: One live inversion and what it buys, 2005:07–2008:12. (a) The Australian 10Y−2Y slope. The naive inversion is the grey bar (slope below zero); the live period under the paper's rule is shaded, from the month after the inversion is observed (▼) to the release (×) after two consecutive steepening months (circles). The curve is released in January 2007 while still inverted and stays naively inverted for another eighteen months. (b) Across the nine curves, the naive inversion count and the IYC live count N_t against the cluster threshold of two. (c) The carry trade's cumulative total return (solid) and its spot component (dashed), with the months in which each traded state is on shaded (states lagged one month). From January to June 2007 the naive state is on and the IYC state is off, and the carry trade earns +9.7%; in October 2008 the naive state is off (the count had fallen to one in September and falls to zero in October) while the IYC state is still on, and the carry trade loses 11.6%.

2.3 Delayed compensation and the fresh-cluster prediction

The crash, the kernel, and the one friction that does the work complete the primitives.

Assumption 3 (Crash and carry)**.** *The high-rate basket suffers a jump of size $J > 0$ only in state R, with intensity $\lambda_t^P = \lambda^R \, \mathbf{1}\{g_t = R\}$. The carry excess return is $rx_{t+1} = f_t - JZ_{t+1} + \eta_{t+1}$, where Z_{t+1} is the crash indicator with $\Pr_t(Z_{t+1} = 1) = \lambda_t^P$, f_t is the forward premium (the rate differential, by covered interest parity), and η is mean-zero noise.*

Assumption 4 (Kernel and intermediary)**.** *The pricing kernel $M > 0$ is high in R, and the marginal carry investor is a capacity-constrained intermediary (Gabaix and Maggiori, 2015).*

Assumption 5 (The named friction: a predetermined forward)**.** *The traded differential decomposes as*

$$f_t \;=\; c_t \;+\; \hat{\Lambda}\,\lambda^R\,\mathbb{E}_{t-1}[p_t]\,J, \tag{10}$$

where $c_t \geq 0$ is its slow-moving level component (the inflation and neutral-rate differentials, taken as given) and the second term is crash compensation marked to the lagged *belief, with p_t the filtered recession probability and $\hat{\Lambda} \geq 1$ a disaster markup.*

Assumption 5 is the single economic commitment that does the work. Infrequent portfolio adjustment (Bacchetta and van Wincoop, 2010) and intermediary capacity (Gabaix and Maggiori, 2015) each deliver a forward that absorbs hazard news with a delay; the assumption writes either as a one-period lag in the crash term. Two features matter below. Because f tracks the hazard with a delay rather than ignoring it, the mispricing opens at the *onset* of a scare and closes as f catches up, which is why freshness matters. And the level component c_t, which exists for reasons outside a disaster model, is what makes carry profitable outside scares and what makes the compensation state-dependent inside them.

Proposition 1 (Insurance ratio at least one)**.** *With correct beliefs and no frictions, the Euler* *equation* $\mathbb{E}_t[M\, rx_{t+1}] = 0$ *implies*

$$\kappa_t^{fric} \;=\; \frac{f_t}{\lambda_t^P J} \;=\; \frac{\mathbb{E}_t[M \mid crash]}{\mathbb{E}_t[M]} \;\equiv\; \Lambda_t \;\geq\; 1, \tag{11}$$

strictly so whenever the crash concentrates in the high-M state.

Status: derived. Because the crash strikes in the high-marginal-utility state, carry *over*-compensates a correctly perceived crash (Rietz, 1988; Barro, 2006). Within this benchmark, a correctly perceived crash covariance therefore cannot produce the conditional loss that Section 5 estimates; reproducing it requires the friction of Assumption 5.

Let $P^\star$ denote the recession posterior conditional on a fresh synchronized onset. Substituting the predetermined forward into the insurance ratio gives the central result.

Proposition 2 (Conditional loss at the fresh synchronized onset)**.** *Conditional on a fresh, synchronized ($k_t \geq 2$) inversion onset from the anchored prior ($\mathbb{E}_{t-1}[p_t] \approx q_0$), define the un-priced hazard $c^\star \equiv \lambda^R J\,(P^\star - \hat{\Lambda} q_0) > 0$. Then*

$$\kappa_t \;=\; \frac{c_t + \hat{\Lambda}\lambda^R q_0 J}{\lambda^R P^\star J} \;<\; 1 \qquad \Longleftrightarrow \qquad c_t \;<\; c^\star, \tag{12}$$

and the predictable one-period excess return is $\mathbb{E}_t[rx_{t+1}] = c_t - c^\star$*, negative in the same event.*

Status: derived given Assumption 5; the assumption is a candidate mechanism, not an identified one (Section 2.6). Two implications carry to the empirics. The onset must be *fresh*, because the un-priced hazard is proportional to the belief innovation—maximal at the onset, zero for a standing inversion whose information f has already absorbed—which is the signal's entry rule. And the same onset arrives with $\kappa_t \geq 1$ whenever the level component of the differential is large, $c_t \geq c^\star$, the *compensated* case that Proposition 4 turns into the overwrite. On the count itself the model adds one statement, proved in Appendix B: under Assumption 2 the count k_t is sufficient for the common state within the binary observation set (Lemma 1 there), so conditioning on the count discards nothing that the pattern of inversions contains; the placement of the threshold at two is the empirical matter of Section 2.2.

2.4 The insurance ratio and conditional UIP

Whether a hedge is worth it at a signal onset turns on one comparison, the reduced-form counterpart of the insurance ratio in Propositions 1 and 2. With crash probability λ_t and crash size J,

$$\mathbb{E}_t\left[r^c_{t+1}\,\middle|\,\text{signal}\right] \;=\; C_t - \lambda_t J \;\lesseqgtr\; 0 \quad\iff\quad \kappa_t \;\equiv\; \frac{C_t}{\lambda_t J} \;\lesseqgtr\; 1, \tag{13}$$

so an unwind or an anti-carry position raises expected return if and only if $\kappa_t < 1$, an accounting statement rather than a model result. The decision is two-sided. Staying invested bears the crash; hedging bears the forgone compensation, and an anti-carry position bears it twice when no crash comes. The question a trader faces at a signal onset is therefore not whether λ_t has risen but whether C_t is large enough relative to $\lambda_t J$, and the carry trade is unusually suited to the question because C_t is observable at t while $\lambda_t J$ is not. The curve informs both terms. The count of Section 2.2 is a statistic of λ_t; the differential and, as Section 2.5 argues, the shape of the inverted curves are information about κ_t.

The same ratio has a regression image in the oldest equation of the field.

Proposition 3 (Conditional Fama slopes)**.** *With crash exposure aligned to the differential across currencies ($J_i \propto f_i$, the trait alignment of Section 2.1), (i) conditional on no-signal months the population Fama slope is $\beta_F^{\mathit{off}} \approx 0$, far above the UIP benchmark of -1, because $f_i > 0$ while $\lambda^P \approx 0$ implies no systematic depreciation; and (ii) conditional on signal months,*

$$\beta_F^{on} \;=\; -\frac{1}{\kappa}, \tag{14}$$

below the UIP benchmark exactly when the conditional loss exists ($\kappa < 1$) and equal to -1 at fair pricing.

Status: derived (Appendix B). The slopes estimated in Section 4 below are +0.50 outside and −1.33 inside, consistent with both signs, and the implied $\hat{\kappa} = 1/1.33 \approx 0.75$, an average across all signal months, will give an independent estimate of the friction's size. The proposition takes no stand on the source of the unconditional premium; it says that whatever that source, a state with an expected carry loss must show a slope below −1, and that the signal is where such a state is observable.

2.5 Hedging, curve shape, and the overwrite

A carry trader who observes the state can protect the portfolio in three ways; the payoff of each, against staying invested at $r^c = C + X$, is

$$\begin{aligned} \text{option hedge:} \quad & C_t + X_{t+1} + \max\{-X_{t+1} - K,\, 0\} - \pi_t, \\ \text{unwind:} \quad & 0, \\ \text{anti-carry:} \quad & -\,(C_t + X_{t+1}). \end{aligned} \tag{15}$$

Buying out-of-the-money puts on the long leg prices the tail explicitly at a premium π_t, and how much of the differential that premium consumes is the subject of Burnside et al. (2011) and Jurek (2014). This paper studies the other two hedges, which need no option market and are what most carry portfolios actually do (the paper uses *unwind* and *exit* interchangeably, and calls the reversed position *anti-carry*). Their costs are asymmetric. Unwinding forgoes $C_t + X_{t+1}$ for as long as the portfolio is flat, a cost that accumulates with the length D of the unwind, $\sum_{d=1}^{D}(C_{t+d-1} + X_{t+d})$. Going anti-carry earns the crash if it comes but loses $2\,(C_t + X_{t+1})$ if it does not, because the short position pays the differential and loses the rally.

In a growth scare the differential is small and the expected crash large, $\kappa_t < 1$, and unwinding is right. But there are states—high carry, high inflation, high current rates—in which the compensation is high and the risk correspondingly lower, $\kappa_t \geq 1$. A depreciation can still occur, but the differential covers it on average, and a hedge forgoes it. Call this the *carry-premium risk* of a hedge, the forgone compensation when the portfolio is out or anti-carry in a state where the differential was adequate. It is the mirror image of crash risk, it grows with the length of the unwind, and it is largest precisely where the differential is largest. Two disciplines follow for any hedging rule. On *duration*, re-enter once the worst of the risk is priced rather than staying out through the recovery; that is the exit rule of Section 2.2. On *state*, do not unwind when the inversion is a normalization from a policy peak rather than a growth scare; the compensation-regime discussion below gives that state an observable. The same differential-versus-crash logic applies to emerging-market carry, where it predicts—and Section 7 finds—no such carve-out, because the EM differential in that regime is not high relative to its own norm.

The compensation regime. Inversion says the expected path falls; the shape of the curve says why, and the why determines κ_t. Under the expectations hypothesis each yield is an average of the expected policy path, $y_t^{(n)} \approx \frac{1}{n}\sum_{h<n} \mathbb{E}_t[r_{t+h}] + \tau$, so a curve that slopes downward along its entire length—three-month rates highest, ten-year lowest—is consistent only with an expected path whose averages fall at every horizon, a normalization from a peak, the inflation-fighting inversion of Section 2.2. The paper calls this configuration *downcurve* and defines it on the average G10 curve, which is built tenor by tenor as the cross-sectional average of the nine countries' yields,

$$\bar{y}_t^{(n)} \;=\; \frac{1}{9}\sum_{i=1}^{9} y_{i,t}^{(n)}, \qquad n \in \{3M,\, 2Y,\, 10Y\}, \tag{16}$$

so that the regime is

$$D_t = \mathbf{1}\{\bar{y}_t^{3M} > \bar{y}_t^{2Y} > \bar{y}_t^{10Y}\}, \tag{17}$$

lagged one month like the signal. Inversion is a statement about one pair of tenors on each curve; downcurve is a statement about the whole shape of the average curve, and it adds

Table 1: Stylized facts by state. Panel A reports months, the lagged average G10 three-month yield and money-market rate, the lagged U.S. money-market rate, average CPI inflation (nine currencies, year on year, lagged two months; all %), and the carry trade's interest rate differential, spot return, and total return (annualized, %/yr), in the full sample, by downcurve regime, and by IYC signal state. Panel B reports the downcurve gap in the full sample, the gap between signal and no-signal months, and the downcurve gap within signal months, with the wild cluster bootstrap p in brackets and stars by that p (regime runs, signal episodes, and signal runs as the respective clusters; B = 9,999). States lagged one month; sample 1988:01–2026:02. The tail and coverage statistics by state that the text cites (the share of loss months, the spot leg's expected shortfall and worst month, and the differential's coverage of the loss) are computed in the same script and reported in Sections 4 and 5.

		Rates and inflation (%, lagged)				Carry trade (%/yr)		
State	Months	3M yield	G10 rate	U.S. rate	CPI YoY	Differential	Spot	Total
Panel A: means by state								
Full sample	458	3.53	3.65	3.22	2.19	+3.91	+0.27	+4.18
non-downcurve months	406	2.94	3.05	2.71	1.86	+3.69	+0.40	+4.09
downcurve months	52	8.14	8.33	7.19	4.75	+5.62	−0.76	+4.86
No signal	366	2.90	3.02	2.57	1.88	+3.60	+2.85	+6.45
IYC signal	92	6.04	6.19	5.80	3.44	+5.14	−10.00	−4.86
non-downcurve months	57	3.91	4.00	4.55	2.50	+4.31	−15.55	−11.24
downcurve months	35	9.51	9.75	7.85	4.96	+6.48	−0.96	+5.51
Panel B: gaps (wild cluster bootstrap p in brackets)								
Downcurve − other, full sample		+5.21	+5.28	+4.48	+2.88**	+1.93	−1.16	+0.77
		[0.327]	[0.339]	[0.188]	[0.033]	[0.523]	[0.845]	[0.715]
Signal − no signal		+3.14	+3.17	+3.23**	+1.56	+1.54	−12.85***	−11.31***
		[0.184]	[0.190]	[0.040]	[0.104]	[0.260]	[0.001]	[0.003]
Downcurve − other, within signal		+5.60**	+5.74**	+3.29**	+2.46**	+2.16	+14.59*	+16.75*
		[0.021]	[0.022]	[0.013]	[0.017]	[0.487]	[0.067]	[0.053]

the three-month point that the inversion ignores. The regime is on in 52 of the 458 months, 35 of them inside the signal, and the split $S_t \times D_t$ is the object Section 5.3 prices. When the front of the curve is not the highest point, the expected easing starts from an unremarkable level, the signature of a growth scare with a small differential and a crashing currency; the paper therefore uses the simplest possible split, downcurve versus the rest. Figure 3 shows when the regime occurs. The downcurve months form two blocks, 1988–91 and 2023–24 (plus a single month in 2007), the sample's two great inflation normalizations, and the signal's episodes inside them are the ones that Section 5.3 finds to be compensated.

Figure 4 shows the four signal-state × regime cells drawn from the data, and Table 1 puts numbers on the income side. The downcurve months are the high-rate, high-inflation months of the sample whether or not the signal is on. Three-month yields are near 9.5%

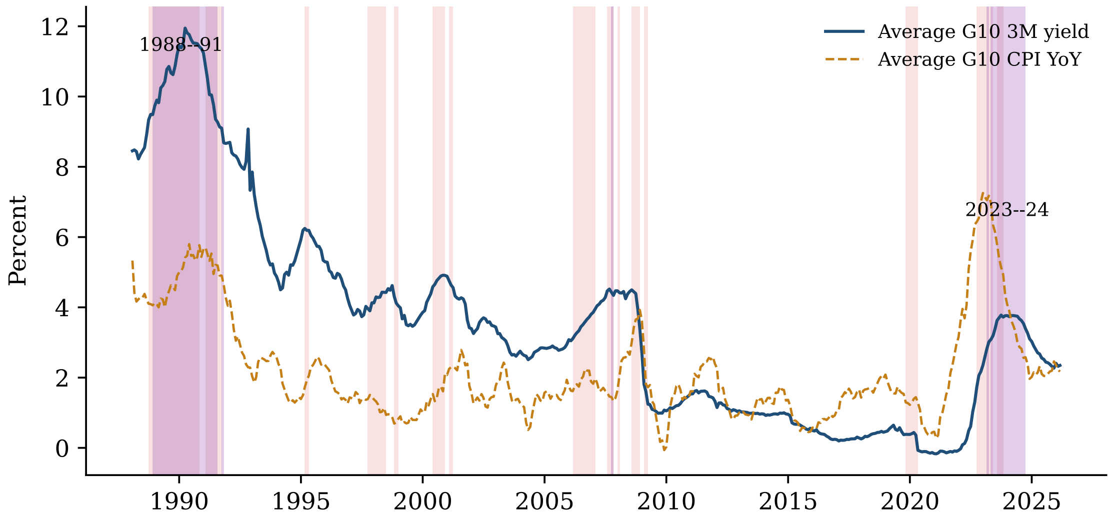

Figure 3: When the downcurve happens. Average G10 three-month yield (solid) and CPI inflation (dashed); purple bands mark downcurve months (D_t of (17) on the average curve), pink bands the IYC signal episodes. The downcurve configuration occurs in the two great inflation normalizations, 1988–91 and 2023–24.

in the downcurve signal months and near 5% in the downcurve no-signal months (mostly the 2023–24 block), against about 4% in the other signal months; across all 52 downcurve months the three-month yield averages 8.1% and inflation 4.8%, against 2.9% and 1.9% elsewhere, and the carry trade's differential is at its largest there (+6.5 against +4.3%/yr within the signal). Panel B puts inference on the gaps. Within the signal the downcurve gaps in rates and inflation are significant at the five-percent level once signal runs are the unit of inference; over the full sample only the inflation gap is, because the 52 downcurve months form three blocks and leave few clusters; the differential gap is positive in every split but not significant; and of the signal-versus-no-signal gaps in rates and inflation only the U.S. rate's is significant (five percent), the inflation gap just missing the ten-percent level ($p = 0.104$). The return consequence of the split is Section 5.3's.

The hypothesis is about compensation through both terms of κ_t. The numerator is the differential, which Table 1 shows at its highest in the downcurve configuration. The denominator is lower there too, because the crash in (5) is a growth disappointment and an inflation-fighting inversion is expected easing without one, and because the currency is

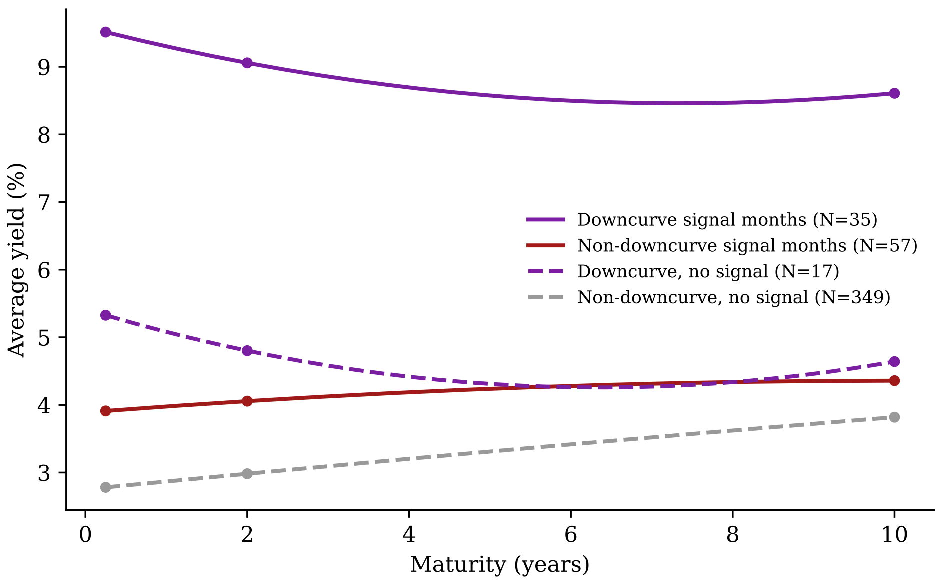

Figure 4: Average G10 yield curves by signal state and downcurve regime (the regime D_t of (17), lagged one month). Solid lines are downcurve and non-downcurve signal months, dashed lines downcurve and non-downcurve no-signal months. Points are the average 3-month, 2-year, and 10-year yields. The downcurve configuration sits at the higher rate levels in either signal state—the regime is an environment, not a byproduct of the signal—and is the compensated cell.

defended by the highest yields on offer, the inflation-fighting support channel of Clarida and Waldman (2008). Bearing the signal's risk should therefore be *paid* in the downcurve regime and *unpaid* in the rest. Section 5.3 reports that this is the pattern, and that the difference is large and borderline at conventional levels (the downcurve wedge inside signal months is +16.8%/yr, $t_{NW} = 3.2$, wild $p = 0.05$); it also records that the regime was located by asking where the signal's false alarms sat, so that the non-signal downcurve months of 2023–24 are its closest out-of-sample check. The portfolio implication has a name in practice, a carry *overwrite*. When the cluster arrives in the downcurve configuration, $\kappa_t \geq 1$, exiting is at its most expensive, and the overwrite stays in. It is the same signal read from the income side.

The overwrite rule.

Proposition 4 (Overwrite rule)**.** *For a risk-neutral one-period position with no transaction costs, the expected-return-maximizing rule at a signal onset is to unwind (or go anti-carry)* *if and only if $\kappa_t < 1$—equivalently, if and only if $c_t < c^\star$. Since κ_t is increasing in c_t, there is a threshold above which the position is retained through the signal.*

Status: derived (immediate from Proposition 2); a risk-averse trader's retention threshold is higher still, so the rule errs toward unwinding. The rule says to stay in through an inflation-fighting inversion and to step aside in a growth scare, the two cases the compensation-regime discussion above makes observable in real time. The overwrite is not a second signal; it is the break-even crossing of the first one.

2.6 Testable implications

The framework leaves five statements for the data, each tested where noted. A fresh, synchronized onset predicts a conditional carry loss, front-loaded and concentrated in the exchange-rate leg (Sections 4 and 5). The loss is ordered by growth exposure and vanishes at zero beta (Section 4). Inside the signal the Fama slope sits below -1 by the size of the conditional loss (Section 4). When the inversion arrives from a high-rate, downward-sloping configuration the position is compensated and the overwrite outperforms the plain exit (Section 5). And the count subsumes the continuous slope, the U.S. curve, and the countries' own signals (Section 6).

Three limits, stated plainly. First, *two frictions are not separately identified.* Pure Bayesian learning delivers a loss only conditional on the latent state; the observable conditional loss requires the predetermined forward, whose degree is measured—not assumed—by how far conditional performance falls below the frictionless benchmark. Second, the count's sufficiency rests on conditional independence and identical loadings across curves. A common term-premium factor threatens the first (Wright, 2011; Hamilton and Kim, 2002); Appendix K confronts it, and the inversions and the signal's losses survive. Unequal loadings would break the second, in which case no linear average of slopes is sufficient either and the count of threshold crossings is the robust aggregator; consistent with that reading, the continuous average slope carries no information for the carry portfolio once the count is controlled for (Section 6). Third, the model shares the empirics' *power ceiling*, since the conditional loss is identified by a handful of delivered scares and the model adds no data. Its contribution is discipline, not proof.

3 Data and Signal Construction

3.1 Data

The sample is monthly, January 1988 to February 2026. As in the portfolio construction of Section 2.1, every return is measured in dollars, the dollar is the funding numeraire and is not itself sorted, and the U.S. curve is not one of the curves the signal counts; Appendix F repeats the headline results with the dollar included in the portfolio and the U.S. curve in the count. The developed-market universe is the nine non-dollar G10 currencies, the Australian dollar (AU), Canadian dollar (CA), Swiss franc (CH), euro (EU; Deutsche mark before 1999), British pound (GB), Japanese yen (JP), Norwegian krone (NO), New Zealand dollar (NZ), and Swedish krona (SE); "G10" hereafter refers to this set. For each country I use end-of-month spot exchange rates, one-month money-market rates, and government yields at three maturities (3 months, 2 years, 10 years), so that the monthly currency excess return decomposes into an income leg (the lagged money-market differential against the U.S.) and a spot leg (the log exchange-rate change); tables state which is used. Currency betas are estimated against the S&P 500 monthly total return, the baseline market index, and against the MSCI World index in the robustness of Appendix G; the S&P 500 return is also the equity outcome of Section 9. Risk controls are the Chicago Fed NFCI (level and one-month change; the series exists from 1971, so the controlled regressions cover the whole sample) and the VIX (level and one-month change; from 1990, the index's inception). The emerging-market universe of Sections 7–8 is the seven floating EM currencies with complete spot and money-market series, the Brazilian real, Chilean peso, Indonesian rupiah, Indian rupee, Mexican peso, Korean won, and South African rand, observed as end-of-month spot rates against the dollar and one-month money-market rates from January 1996 (the Chilean series is the binding start), together with their 2-year and 10-year government yields, which start at staggered dates and enter only the EM-native check of Appendix I. G10 inflation is the equal-weighted average of the nine countries' year-on-year CPI inflation, monthly for seven of them and quarterly for Australia and New Zealand, whose latest published quarter is carried forward; it is used only to characterize eras and regimes. G10 activity, used in the local projections of Section 4, is a proxy because the full set of nine countries is not available at a monthly frequency. It is industrial production for the seven countries that publish it monthly (Canada, Switzerland, Germany, the United Kingdom, Japan, Norway, and Sweden; Australia and New Zealand publish it quarterly) and the OECD composite leading indicator for the five countries the OECD covers (Australia, Canada, Germany, Japan, and the United Kingdom), both from the OECD. Appendix A details coverage.

3.2 The IYC signal

The IYC signal is the construction of Section 2.2, built on 10Y−2Y inversions and lagged one month, the trader's timing convention. The signal and every conditioning variable are measured at the end of a month, the position decision—stay, unwind, re-enter—applies to the following month, and every regression is therefore predictive, next month's return on this month's state. In the tables' notation the state carries the subscript $t-1$ against a month-t return; nothing dated t enters the information set that positions for month t (the lag audit is in Appendix E).

The naive count—at least two of the nine curves inverted, with no further discipline—is the benchmark throughout. It identifies the same seasons but is on for 165 months against the signal's 92 (all of 1988–92 and 2022:08–2024:10 included), so its state is mostly waiting, and it is off in the three worst carry months of the sample, which fall in the delivery phase that the latch of Section 2.2 is built to hold. Appendix D contains the full anatomy, including the result that no repair of the naive count—weighting by age, or filtering by the paper's own downcurve regime—reaches the months the discipline finds; Appendix C tabulates the confirmation-length and cluster-size grids, on which the alternatives degrade in the directions the text's arguments predict.

The signal is on in 92 of 458 months, in fifteen episodes (fourteen of them multi-month). The heterogeneity that Section 5.3 prices is already visible. The episodes spent in the downcurve regime—1991, 2023:07–09, and the 1988–90 episode, twenty-three of whose twenty-five months are downcurve—end with positive carry totals, while the growth-scare episodes (1998, 2019–20) and the delivery phases of 1995 and 2008 carry the disasters.

3.3 The sample's economic regimes

The thirty-eight years span very different nominal environments, and the paper's premise—that carry's compensation varies over time—is visible in the rawest era accounting, which Appendix A reports (Table A.2) together with the nine currencies' summary statistics (Table A.3). Two features organize what follows. The carry trade's interest rate differential ranges from +0.9 to +5.9%/yr across eras, a sixfold range, so the compensation is an era variable rather than a constant premium, consistent with the evidence that carry-trade behavior is regime-dependent (Clarida et al., 2009; Christiansen et al., 2011); and the downcurve months sit almost entirely inside the two high-nominal eras (33 of 52 in 1988–94, 18 in 2022–26), so the regime that Section 5.3 shows to be the compensated one is the high-inflation environment in both of its sample appearances.

3.4 Inference

Signal months are serially dependent—episodes are long—so inference must respect the episode structure, and the paper uses one convention throughout. Every reported coefficient carries up to three statistics. The *Newey–West t* (12 lags, denoted t_{NW}) is the standard of the currency literature (Lustig et al., 2011; Menkhoff et al., 2012; Lettau et al., 2014) and treats the regressor's persistence through the HAC bandwidth. The *wild cluster bootstrap* p (Cameron et al., 2008; MacKinnon et al., 2023) (Rademacher weights, null imposed, entire contiguous signal runs resampled, controls partialled under the null; all p-values two-sided), shown in brackets, is the conservative complement. With fifteen episodes it asks whether the *episodes*, not the months inside them, are informative. Where noted, the *episode-clustered* t (t_{ep}) is the plug-in version of the same idea. Continuous regressors (the global slope and its change) have no episode structure and carry t_{NW} only. Significance stars throughout the paper denote * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$—by the wild cluster bootstrap wherever a coefficient reports one, and by the Newey–West t otherwise. Stars follow the exact bootstrap p, while the displayed p-values are rounded to three decimals, so a coefficient can carry three stars beside a displayed 0.010. Wherever other regressors enter a specification (the horse races of Section 6, the control sets of Section 10, and the appendix batteries), they are partialled under the null within the bootstrap.

4 Mechanism: High Rates Are a Growth Exposure

Why does the crash live in the carry trade's long leg? The paper's answer is that a persistently high interest rate is the price of a persistent exposure to global growth. Commodity exporters and small open economies pay high rates *because* they are long the world cycle, and funding havens pay low rates because they are short it. The literature offers complementary reasons the same economies pay persistently high rates—country size (Hassan, 2013), position in the trade network (Richmond, 2019), commodity production (Ready et al., 2017)—all of which load the same currencies on world risk. Five facts establish the chain.

The first fact is that the rate ranking is the growth-exposure ranking. Table 2 puts the mechanism in one exhibit, listing for each currency its average rate differential, its (real-time, expanding-window) equity beta, and its individual on-signal spot effect. The rate

Table 2: The mechanism currency by currency, sorted by beta. Columns: sample-average money-market differential vs. the U.S. (%); sample-average real-time (expanding-window) equity beta; and the currency's own on-state effect b from $y_{i,t} = a_i + b_i\,\mathbf{1}\{\text{state}\}_{t-1} + e_{i,t}$ (annualized, %/yr), where y is the spot return $\Delta s_{i,t}$ or the total excess return $rx_{i,t}$ of (1), for the IYC signal, for its non-downcurve months, and for its downcurve months; no controls, 1988:01–2026:02. The wild bootstrap p on the state's episodes is in brackets beneath each coefficient, and stars follow it.

			IYC signal		Non-downcurve months		Downcurve months	
Currency	Rate diff.	Beta	Spot	Total	Spot	Total	Spot	Total
AU	+1.87	+0.20	−13.21**	−12.34*	−19.28**	−20.50**	−0.28	+3.58
			[0.013]	[0.090]	[0.024]	[0.019]	[0.949]	[0.497]
CA	+0.46	+0.14	−4.61	−4.04	−8.94***	−9.65***	+3.32	+5.73
			[0.220]	[0.360]	[0.006]	[0.006]	[0.331]	[0.248]
NZ	+2.29	+0.13	−13.28**	−12.64**	−18.90**	−19.27**	−1.02	+0.99
			[0.012]	[0.031]	[0.037]	[0.031]	[0.774]	[0.776]
NO	+1.52	+0.07	−3.96	−4.46	−4.91	−6.22	−1.43	−0.54
			[0.539]	[0.486]	[0.575]	[0.434]	[0.875]	[0.954]
SE	+0.62	+0.06	−4.94	−5.11	−7.86	−9.59	+0.91	+3.18
			[0.421]	[0.430]	[0.354]	[0.228]	[0.892]	[0.639]
EU	−0.21	+0.02	−1.19	−2.12	−0.78	−1.56	−1.50	−2.41
			[0.831]	[0.704]	[0.915]	[0.827]	[0.814]	[0.765]
GB	+1.16	+0.02	−3.00	−1.75	−1.46	−2.01	−4.56	−0.88
			[0.563]	[0.764]	[0.836]	[0.775]	[0.648]	[0.897]
JP	−2.18	−0.02	+2.35	+0.91	+8.07	+6.17	−7.11**	−7.46*
			[0.570]	[0.855]	[0.249]	[0.390]	[0.036]	[0.063]
CH	−1.65	−0.03	−0.89	−1.72	+0.56	−0.95	−2.88	−2.45
			[0.868]	[0.746]	[0.912]	[0.850]	[0.742]	[0.734]

Rank corr. (rate diff., beta) = 0.72. Fit of the spot effect on beta—IYC signal: slope = −60.0, $R^2 = 0.77$, intercept = −0.75; non-downcurve months: slope = −105.6, $R^2 = 0.84$, intercept = 1.09; downcurve months: slope = 26.5, $R^2 = 0.47$, intercept = −3.38

and beta rankings coincide (rank correlation 0.72). The persistent high-rate currencies, Australia and New Zealand, are the high-beta currencies, and the funding currencies, Japan and Switzerland, have betas of zero. The high-beta portfolio of Section 2.1 is, in essence, the carry portfolio measured from asset prices instead of money markets, and Section 5 shows that it loses on the signal with the same profile as the carry portfolio (−7.5%/yr spot, wild $p = 0.010$; Table 4). The persistence is in membership, not just ranking. Both portfolios hold essentially the same currencies in and out of the signal (Appendix H), so the conditional losses reprice a standing exposure rather than a rotated one.

The second fact is that the loss scales with the exposure and vanishes without it. In the cross-currency fit (Figure 5) the on-signal loss lines up with beta at $R^2 = 0.77$ with a zero intercept. Australia and New Zealand—the two highest-differential currencies, both in the top three by beta—are the only individually significant losers on the full signal (−13.2 and

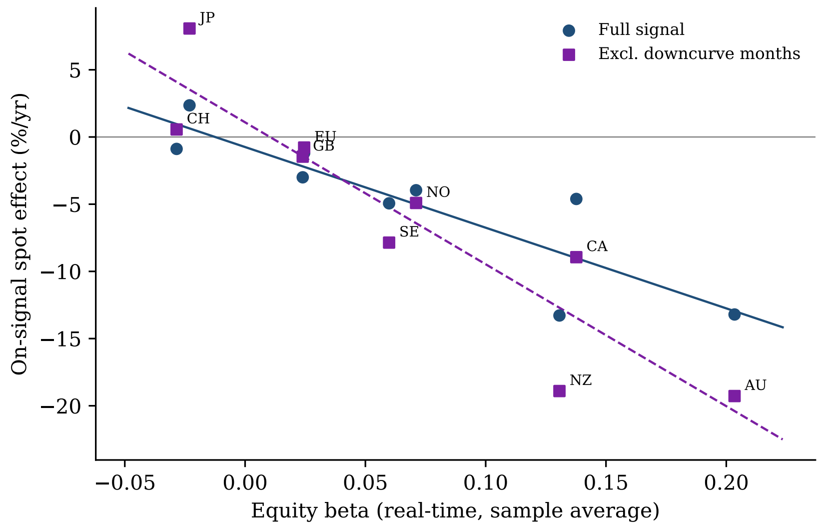

Figure 5: On-signal spot effect against equity beta, one point per currency. Circles and solid fit are for the full signal (slope −60, $R^2 = 0.77$), squares and dashed fit for the non-downcurve months (slope −106, $R^2 = 0.84$). Both intercepts are economically zero, so currencies without growth exposure do not crash on the signal.

−13.3%/yr, wild $p \leq 0.013$), and the zero-beta currencies (Japan, Switzerland, the euro) show nothing ($p \geq 0.57$). The downcurve filter sharpens every part of the picture. On the non-downcurve months the fit steepens to −106 per unit of beta at $R^2 = 0.84$, Australia and New Zealand deepen to −19.3 and −18.9, Canada—the second-highest-beta currency—becomes individually significant (−8.9, $p = 0.006$), and the yen's positive point estimate strengthens to +8.1 (insignificant). The safe haven (Ranaldo and Söderlind, 2010) appreciates in exactly the scares that are unpaid for everyone else. The downcurve columns show the mirror image. In the signal's downcurve months no high-beta currency loses (Australia −0.3, New Zealand −1.0, Canada +3.3, none significant), the cross-currency fit on beta turns flat to positive, and the yen is the one currency with a significant coefficient, a *depreciation* of −7.1%/yr in spot ($p = 0.036$), so the flight to safety that marks a growth scare is absent, indeed reversed, in the inversions the shape classifies as inflation fighting. The total-return columns tell the same story. The interest rate differential does not rescue the high-beta losers—totals track spots nearly one for one (Australia −12.3 against −13.2 on the full signal, though at weaker significance, $p = 0.09$), and on the non-downcurve months the totals are, if anything, slightly larger (Australia, New Zealand, and Canada at $p \leq 0.031$).

The trait, not the channel, is what the signal prices. A third sort on the same family—each currency's *oil beta*, from expanding regressions on WTI returns, the commodity channel of Ready et al. (2017)—replicates the identical anatomy (a significant signal loss, a sharper filtered loss, a flat downcurve stay-set), but it is fully spanned by the two portfolios the paper already holds. Regressed on the carry and high-beta portfolios' returns it has an alpha of -0.4%/yr ($t = -0.6$) with loadings of 0.33 and 0.40 ($R^2 = 0.53$), and its signal cells vanish once those returns are partialled (-8.0 to -1.2 in totals, $p = 0.55$; computations in the replication package). Commodity exposure is one channel of the growth-exposure trait, not a separate factor—which is what "two measurements of the same ranking" predicts. In the pooled panel with episode-clustered inference the signal-times-beta interaction is -27%/yr per unit of beta (smaller than the cross-sectional slope because the panel uses monthly time-varying betas rather than sample averages; unreported) with the signal effect at zero beta indistinguishable from zero (-3.7, $t = -0.8$); the interaction's wild p of 0.21 reflects the fifteen-episode cluster count rather than the cross-section, which is as tight as Figure 5 shows. The cross-sectional address of the loss is the exposure, not the currency label.

The third fact is the oldest expression of the same thing, namely the Fama (1984) regression, run by state. Table 3 regresses monthly appreciation on the lagged differential (UIP benchmark -1) by state. Outside the signal the pooled slope is $+0.50$, the forward-premium puzzle in its classic form, with high-differential currencies failing to depreciate. Inside the signal the implied slope is -1.33, so UIP is not merely restored but overshoots the benchmark. This is what a conditional loss looks like in a Fama regression. If the interest rate differential covers only a fraction κ of the expected loss, the on-signal slope is $-1/\kappa$ (Proposition 3 of Section 2.4), so a slope below -1 is the Fama-regression image of the conditional loss, and the point estimate implies $\hat{\kappa} \approx 0.75$. (The two-state interaction is significant under month-level HAC inference, $t = -2.39$, but not under episode clustering, wild $p = 0.34$; the table reports both.) Splitting by the downcurve regime then shows *where* in the signal the reversal lives, and the answer matches the compensation hypothesis of Section 2.5, which Section 5.3 tests on the portfolio. The entire collapse sits in the non-downcurve months (interaction -5.0, $t_{NW} = -3.7$, wild $p = 0.09$; implied slope -4.6; per-currency slopes negative for eight of nine, the yen the safe-haven exception), while in the *paid* months the puzzle survives untouched (implied slope $+0.4$, interaction ≈ 0, and per-currency slopes positive for six of nine, the yen's and the New Zealand dollar's significantly so). A positive slope is outside the $-1/\kappa$ mapping, which prices a state with a material crash hazard; where the hazard is small—the downcurve months, whose spot tail is a fraction of the other signal months' (expected shortfall -3.7 against -9.0%/month; Section 5)—the state behaves like the off-state and the slope reverts to the unconditional puzzle. UIP breaks down exactly where the differential fails to cover the risk, and continues to fail in the trader's favor exactly where it does. The reading follows the peso logic, in which the Fama puzzle is the *survival-state* face of the crash premium and the signal—refined by the shape—locates the payback states. Lloyd and Marin (2020) document bilateral UIP reversals conditional on own-curve inversions at longer horizons; the version here conditions on the global cluster and its compensation regime.

The fourth fact is that the leading indicator of activity falls after the signal. Figure 6 traces what follows a signal month with local projections of four series on the lagged state, at horizons of one to twelve months, with the wild cluster bootstrap on signal runs as the inference. The activity series are the proxies of Section 3, the currency countries' own industrial production and leading indicator where the OECD publishes them monthly. Measured activity deteriorates. The leading indicator falls by 0.6% over the six months after a signal month (wild p between 0.01 and 0.05 from the first to the seventh month) and industrial production by 1.8% at six months and 2.0% at seven ($p = 0.06$ and 0.05, at or below 0.06 through the tenth month), the indicator recovering about half of its fall by the twelfth month and industrial production little of it. Because the state is persistent, a signal month's twelve-month window overlaps with its run's own continuation, so the same projections are run on

Table 3: Fama regressions by state. Specification: $\Delta s_{i,t+1} = a_i + \beta_F \frac{1}{12}(i_{i,t} - i_{US,t}) + e_{i,t+1}$, monthly appreciation vs. USD (%/month) on the lagged money-market differential (%/month), 1988:01–2026:02. Per-currency slopes with Newey–West (12 lags) t-statistics, estimated separately on the no-signal months, the IYC signal months, and the signal months outside and inside the downcurve regime. Pooled row: country fixed effects, with the differential interacted with the signal (two-state specification) or with the non-downcurve and downcurve signal months (three-state specification); the entry is the implied slope in the state, with the interaction's Newey–West t in parentheses and its wild cluster bootstrap p on signal runs in brackets (B = 9,999), stars by that p. UIP implies a slope of −1.

	No signal		IYC signal		Non-downcurve		Downcurve	
Currency	β_F	t_{NW}	β_F	t_{NW}	β_F	t_{NW}	β_F	t_{NW}
AU	+1.63	+1.62	+0.12	+0.07	−8.49	−1.28	+0.66	+0.71
CA	−0.77	−0.80	+1.40	+1.91	−3.21	−0.62	+0.25	+0.23
CH	+2.00	+1.97	−1.93	−0.79	−6.07	−1.33	+0.94	+0.26
EU	+1.60	+1.20	−5.87	−1.60	−9.02	−1.71	−2.08	−0.51
GB	+0.14	+0.09	−0.62	−0.38	−3.96	−0.68	+1.96	+0.80
JP	+1.47	+1.51	+0.88	+0.97	+0.19	+0.10	+3.60	+3.37
NO	−0.67	−0.56	−3.65	−1.35	−5.35	−1.24	−2.97	−0.96
NZ	+0.64	+0.74	−1.12	−0.36	−6.62	−1.37	+3.02	+3.81
SE	−0.76	−0.54	−1.74	−0.85	−5.94	−0.87	−2.35	−1.17
Pooled, currency fixed effects: implied slope by state (interaction t_{NW} in parentheses, wild p in brackets)								
Pooled	+0.50	(+0.97)	−1.33	(−2.39)	−4.64*	(−3.72)	+0.40	(+0.02)
				[0.340]		[0.088]		[0.994]

the onset month of each run alone (fifteen events) and with the lagged state and three lags of the outcome's monthly change as controls (Appendix J). The indicator's decline is the same under both (−0.5 and −0.6% at six months, p = 0.06 and 0.07). Industrial production is the weaker series. Its decline after an onset is larger (−2.6% at seven to nine months, p from 0.03 to 0.10) but it is not significant once the lagged state and its own recent changes are controlled (−1.7% at eight months, p = 0.23), so the activity evidence rests on the leading indicator. The carry spot leg's loss is front-loaded, −1.1% in the first month ($p < 0.01$) and −2.4% cumulative by the fourth (p = 0.06) on the state, and larger on the onset and controlled designs (−3.7 to −4.2% by the fourth to sixth month, $p \leq 0.07$), with a partial rebound after. The average G10 money-market rate, by contrast, does not move on average over the year after a signal month under any of the three designs; the signal pools the growth scares, in which rates are cut, with the inflation normalizations, in which they stay high,

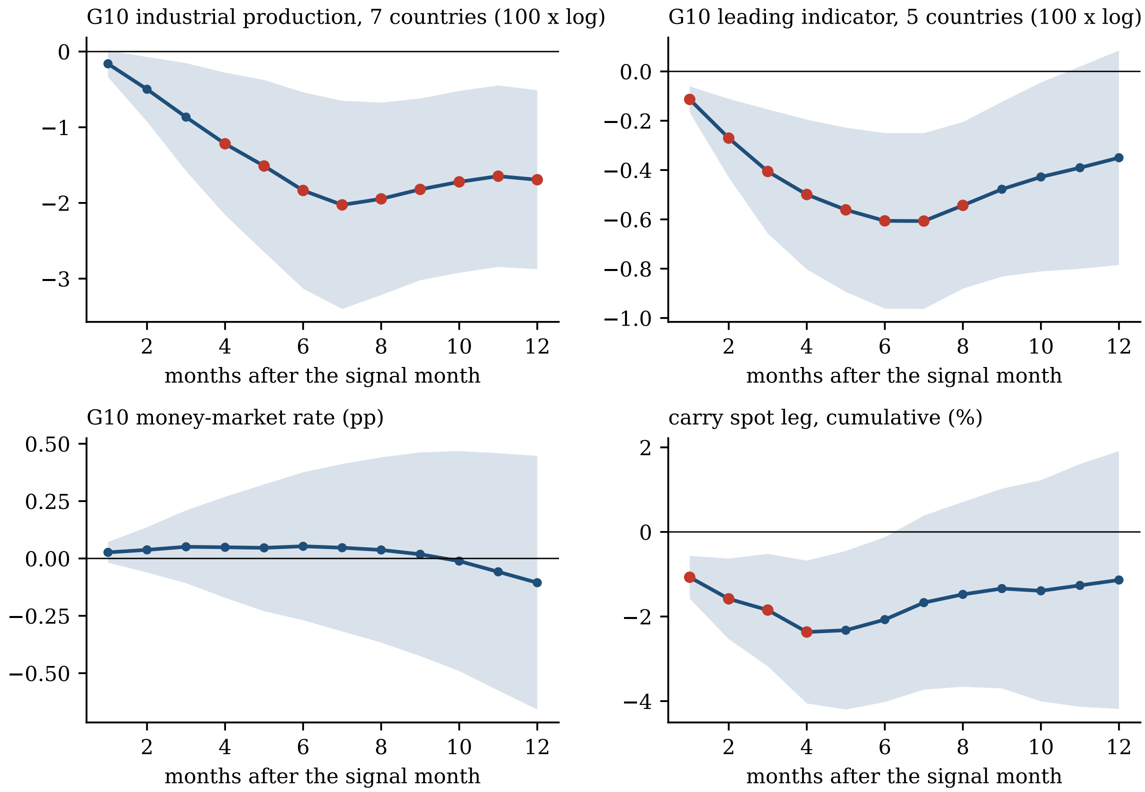

Figure 6: What follows a signal month. Local projections of the cumulative change from $t-1$ to $t-1+h$ on the lagged signal state, for $h = 1$ to 12 months, 1988:01–2026:02, of the G10 activity proxies of Section 3—industrial production for the seven currency countries that publish it monthly and the OECD composite leading indicator for the five it covers, each the cumulated average of the countries' monthly log changes (100 $\times$ log)—the average G10 one-month money-market rate (percentage points), and the carry trade's spot leg (cumulative %). Bands are 90% Newey–West intervals with $h+1$ lags; red markers have a wild cluster bootstrap p below 0.10 with signal runs as clusters. Appendix J repeats the projections on the onset month of each run and with lagged controls.

and Section 5.3 separates the two. Within the signal, the activity decline is concentrated in the non-downcurve months (industrial production -2.7% at six months against -0.5% in downcurve months; the indicator -0.8 against -0.4), although with fifty-seven and thirty-five months the contrast is not significant (wild $p \geq 0.22$). The cluster is doing what the expectations hypothesis says it should—aggregating market-implied easing expectations that anticipate a real slowdown—and the growth-exposure reading is consistent with the direct evidence that currency premia compensate business-cycle risk (Colacito et al., 2020).

The fifth fact is that the delivery phase carries the exchange-rate loss. Within signal episodes the worst months arrive after the naive inversion count has receded, and the episode's currency damage is complete once the re-steepening confirmation arrives (Figure 1; Section 11).

Together these facts discipline the interpretation of the portfolio results that follow. The signal does not identify "carry risk" in the abstract; it identifies moments when the global growth exposure embedded in high-rate currencies is about to be repriced. The interest rate differential and the crash are the same trait in different states of the world—which is why the downcurve regime of Section 5.3 can price when the differential is large enough to bear it.

5 The Carry Trade In and Out of the Signal

5.1 Conditional moments and state regressions

Table 4 reports the carry trade's conditional distribution in and out of the IYC signal's 92 months, for total returns and the spot leg, with the high-beta portfolio alongside, and splits the signal months by the downcurve regime of Section 2.5. The last column is the state regression $y_t = a + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_t$, with y the annualized monthly return, no controls, and the wild cluster bootstrap p on the state's episodes in brackets; it is the paper's headline specification, the portfolio-level form of the currency-by-currency regression of Table 2, and the controlled and robustness tables build on it.

Outside the signal, carry is the strategy the literature knows, earning +6.5%/yr in total returns at a Sharpe ratio of 0.89. Inside the signal every moment deteriorates at once. The mean swings to −4.9%/yr, the Sharpe ratio to −0.51, the skewness from −0.34 to −0.98, and the expected shortfall from −4.4 to −7.8% per month. The spot leg shows the repricing undiluted (+2.9 to −10.0%/yr), and the high-beta portfolio shows the same swing, as the

Table 4: The carry trade in and out of the IYC signal. For each series and state, the number of months, the annualized mean (%/yr), the annualized Sharpe ratio, the skewness, and the expected shortfall (the mean of the worst 5% of monthly observations, %/month); states are the no-signal months, the IYC signal months, and the signal months outside and inside the downcurve regime, all lagged one month. The last column is the coefficient b on the state dummy in $y_t = a + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_t$ over all months (so the base is every other month), with the wild cluster bootstrap p on the state's episodes in brackets and stars by that p ($B = 9{,}999$). The *ex bottom decile* row re-estimates the signal regression after dropping the series' full-sample bottom-decile months from both states, the ex-crash shift discussed in Section 5. Sample 1988:01–2026:02 ($N = 458$).

Series	State	N	Mean (%/yr)	Sharpe	Skew	$\text{ES}_{5\%}$	b [wild p]
Carry, spot	No signal	366	+2.85	0.39	−0.41	−4.80	
	IYC signal	92	−10.00	−1.05	−0.99	−8.14	-12.85^{***} [0.001]
	non-downcurve	57	−15.55	−1.45	−0.92	−9.02	-18.07^{***} [0.001]
	downcurve	35	−0.96	−0.15	0.33	−3.71	−1.33 [0.671]
	ex bottom decile	77	+0.54	0.09	0.66	−2.50	-7.24^{***} [0.006]
Carry, total	No signal	366	+6.45	0.89	−0.34	−4.44	
	IYC signal	92	−4.86	−0.51	−0.98	−7.77	-11.31^{***} [0.003]
	non-downcurve	57	−11.24	−1.05	−0.92	−8.70	-17.61^{***} [0.002]
	downcurve	35	+5.51	0.83	0.49	−3.00	+1.45 [0.581]
	ex bottom decile	77	+5.82	0.94	0.72	−2.04	-5.50^{**} [0.024]
High-beta, spot	No signal	366	+1.26	0.16	−0.43	−5.05	
	IYC signal	92	−6.27	−0.79	−0.29	−5.33	-7.53^{***} [0.010]
	non-downcurve	57	−11.54	−1.47	−0.51	−5.74	-12.89^{***} [0.000]
	downcurve	35	+2.31	0.31	0.13	−4.35	+2.78 [0.407]
	ex bottom decile	77	+2.55	0.44	0.51	−2.68	-3.72^{*} [0.084]
High-beta, total	No signal	366	+3.18	0.41	−0.43	−4.88	
	IYC signal	92	−3.64	−0.46	−0.27	−5.11	-6.82^{**} [0.013]
	non-downcurve	57	−9.21	−1.17	−0.51	−5.55	-12.59^{***} [0.001]
	downcurve	35	+5.43	0.73	0.21	−4.09	$+3.92^{*}$ [0.084]
	ex bottom decile	78	+4.67	0.80	0.54	−2.63	-3.68^{*} [0.060]

mechanism of Section 4 says it must. In the regression column the carry spot leg loses 12.9%/yr in signal months (wild $p = 0.001$) and total returns 11.3 ($p = 0.003$); the high-beta portfolio loses 7.5 in spot ($p = 0.010$) and 6.8 in totals ($p = 0.013$).

The downcurve split shows where the loss lives. In the signal months outside the downcurve regime every cell sharpens—carry spot −18.1%/yr ($p = 0.001$), totals −17.6, and the high-beta portfolio −12.9 and −12.6, all at $p \leq 0.002$—while the downcurve signal months show no loss in total returns and no significant loss in any cell. Their carry total is +5.5%/yr at a Sharpe ratio of 0.83 with positive skewness, their expected shortfall is smaller than the

Table 5: The conditional distribution under the signal. Quantiles of monthly returns (%/month; spot and total for both portfolios), annualized volatility, and skewness, out of the signal, in its 92 months, and in the signal months outside and inside the downcurve regime; 1988:01–2026:02. Supporting statistics in the text: variance-ratio permutation p-values resample contiguous signal episodes; ex-crash regressions drop each portfolio's full-sample bottom-decile months from both states; hazard regressions are linear-probability models of the crash indicator on the lagged signal with episode-clustered wild bootstrap p.

Series	State	N	q10	q25	q50	q75	Vol (ann.)	Skew
Carry, spot	Off	366	−2.35	−0.93	0.35	1.52	7.23	−0.41
	On	92	−3.94	−1.98	−0.76	0.94	9.55	−0.99
	On, non-downcurve	57	−5.27	−2.56	−1.04	0.84	10.73	−0.92
	On, downcurve	35	−1.72	−1.43	−0.17	0.98	6.60	0.33
Carry, total	Off	366	−2.03	−0.62	0.67	1.88	7.27	−0.34
	On	92	−3.59	−1.49	−0.17	1.39	9.61	−0.98
	On, non-downcurve	57	−4.99	−2.18	−0.64	1.19	10.69	−0.92
	On, downcurve	35	−1.39	−0.95	0.27	1.55	6.62	0.49
High-beta, spot	Off	366	−2.69	−1.19	0.25	1.59	7.75	−0.43
	On	92	−3.80	−1.72	−0.36	0.93	7.94	−0.29
	On, non-downcurve	57	−4.39	−2.69	−0.42	0.75	7.87	−0.51
	On, downcurve	35	−2.06	−1.04	−0.13	1.70	7.51	0.13
High-beta, total	Off	366	−2.58	−1.03	0.41	1.73	7.77	−0.43
	On	92	−3.62	−1.49	−0.13	1.16	7.95	−0.27
	On, non-downcurve	57	−4.22	−2.48	−0.31	0.94	7.87	−0.51
	On, downcurve	35	−1.70	−0.64	0.18	1.85	7.47	0.21

no-signal months', and every coefficient is positive or insignificant. The differential is never negative in signal months, so an exit always sacrifices income; the split identifies the signal months in which the differential is large enough to cover the spot risk it accompanies. Removing those months does not dilute the crash signal, it concentrates it, because they were never the risk.

5.2 Shift or tail? Both, and the division of labor is informative

Table 5 asks what kind of conditional change the signal induces, a shift of the whole distribution or a fattening of the tail.

The answer is both, with an informative division of labor between the two portfolios. For the *carry portfolio*, the signal shifts the entire distribution, and this is the fact that answers the reverse-engineering objection to the latch. Every quantile moves left (the median from

+0.35 to −0.76%/month), and the shift survives excluding crash months altogether. The *ex bottom decile* rows of Table 4 drop each series' full-sample bottom-decile months—all 46 worst months, including the three the latch is built to keep—from both states, and the carry portfolio's remaining signal-month deficit is still −7.2%/yr in spot (wild $p = 0.006$) and −5.5 in total ($p = 0.024$). A rule reverse-engineered around three crashes would show nothing here. For the *high-beta portfolio*, the signal is a cleaner tail state, in which the probability of a crash month roughly doubles (8.5% to 16.3% at the bottom-decile definition, wild $p = 0.046$; 4.6% to 9.8% at a −4%/month cutoff, $p = 0.033$) with essentially no change in variance (ratio 1.05, permutation $p = 0.80$) and an ex-crash shift about half the carry portfolio's and marginal (−3.7%/yr in spot, $p = 0.084$; −3.7 in total, $p = 0.060$)—the carry result is the strong one, and the table reports both. Crash severity, conditional on occurring, is roughly constant in both portfolios (−4.2 to −5.3%/month in and out of the state). The carry portfolio's in-state variance is higher in point estimate (ratio 1.74) but not significantly so ($p = 0.14$); the controlled regressions of Section 10 give the sharper answer to the volatility question, since conditioning on the VIX and financial-conditions gauges strengthens rather than weakens the signal. The downcurve rows show where the tail sits inside the signal. The non-downcurve months carry the left tail of both portfolios (the carry portfolio's tenth percentile is −5.3%/month there against −2.4 outside the signal), while the downcurve months' lower tail is thinner than the no-signal months' (−1.7 and −1.4 for spot and total) even though their center sits somewhat below it, the distribution of a state in which the spot leg drags and the differential covers the drag.

5.3 Hedging: unwind, anti-carry, and the carry overwrite

Table 6 evaluates the signal where a portfolio manager would, namely as an overlay on the carry trade's total returns, against the always-invested baseline; Figure 7 plots the cumulative returns.

When the signal fires, the manager has the two hedges of Section 2.5, and the decision

Table 6: Trading the signal. The same five strategies applied to the carry trade and to the high-beta portfolio (total returns, 1988–2026, no transaction costs). The overwrite strategies stay invested when the average curve is in the downcurve configuration, where the carry premium covers the risk.

Strategy	Mean	Sharpe	Skew	$ES_{5\%}$	Max DD	Worst mo.
Panel: the carry trade (total returns)						
Always in	+4.18	0.53	−0.69	−5.32	−30.1	−11.61
Exit on signal	+5.15	0.79	−0.21	−4.25	−25.9	−5.78
Anti-carry on signal	+6.13	0.79	+0.12	−4.50	−25.9	−5.78
Overwrite: exit unless downcurve	+5.57	0.82	−0.21	−4.29	−25.9	−5.78
Overwrite: anti-carry unless downcurve	+6.97	0.90	+0.15	−4.38	−25.9	−5.78
Panel: the high-beta portfolio (total returns)						
Always in	+1.81	0.23	−0.40	−5.03	−25.8	−8.97
Exit on signal	+2.54	0.37	−0.40	−4.66	−24.7	−8.97
Anti-carry on signal	+3.27	0.42	−0.29	−4.80	−35.0	−8.97
Overwrite: exit unless downcurve	+2.96	0.41	−0.36	−4.71	−20.7	−8.97
Overwrite: anti-carry unless downcurve	+4.10	0.53	−0.27	−4.71	−21.2	−8.97

is two-sided, the risk of remaining invested against the carry-premium risk of hedging. The costs of the two hedges are asymmetric. An unwind that is wrong forgoes the differential and the rally; an anti-carry position that is wrong pays the differential and loses the rally, double the cost. The carry panel prices the trade-off. Exiting raises the mean from +4.2 to +5.2%/yr and the Sharpe ratio from 0.53 to 0.79, halving the worst monthly loss (−11.6 to −5.8) and improving the skewness from −0.69 to −0.21; anti-carry earns more (+6.1%/yr, skewness turning positive) but at the same Sharpe ratio—the extra return is bought with doubled exposure to classification error. Both sides of the design contribute to either option, the fresh-cluster entry because it is out (or short) only when that pays, and the re-entry confirmation because it restores the position for the recovery rather than leaving it stranded, the failure mode of the naive count (Section 11). The gate trades only at episode boundaries—thirty position changes in thirty-eight years—so transaction costs do not change the picture. Charging each unit of position change 10, 25, or 50 basis points of the portfolio's notional (an exit or re-entry moves the position by one unit, an anti-carry switch by two) takes the exit's Sharpe ratio from 0.79 to 0.78, 0.76, and 0.73, and the overwrite anti-carry's from 0.90 to 0.88, 0.85, and 0.80, against 0.53 for the always-invested position. The asymmetry

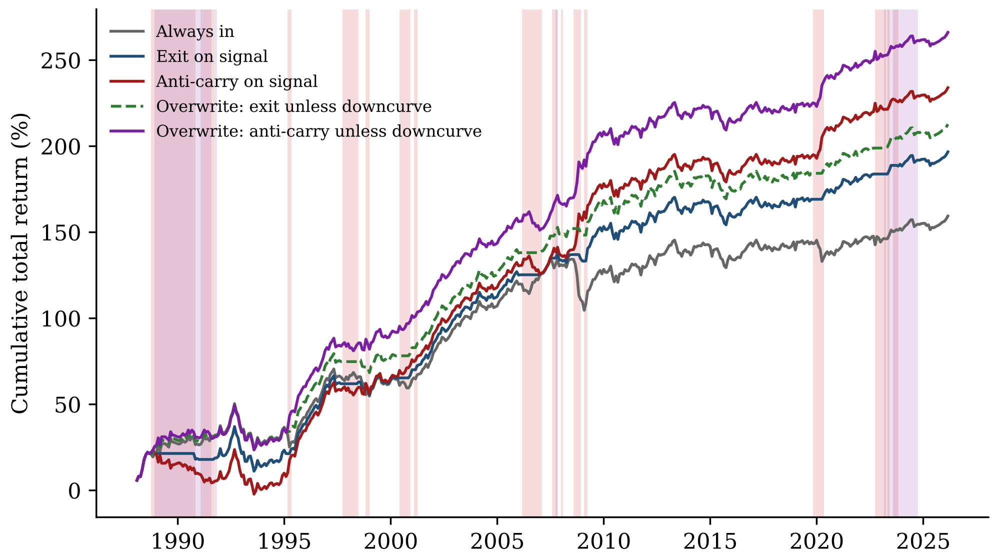

Figure 7: Cumulative total returns of the strategies in Table 6. Pink bands mark the IYC signal episodes, in which the exit and anti-carry strategies depart from the always-invested line, and purple bands the downcurve months, in which the overwrite strategies differ from the plain exit and anti-carry, most visibly in 1988–91 and 2023–24.

between the two options is what the downcurve split of Table 4 resolves.

The diagnostic that located those months is part of the evidence. The signal months in which carry did *not* lose are not scattered randomly across episodes—they cluster in specific periods, and those periods share an environment, the downcurve months of Figure 3, the sample's two great inflation normalizations. Section 2.5 says why these months should be different, the downcurve configuration being the high-level, high-differential one.

Table 1 gives the accounting, downcurve versus all other months in the full sample and inside signal months; the tail-and-coverage numbers cited below are computed by state in the same script. Over the full sample the regime does not predict returns; it changes their composition, replacing a small differential with flat spot by a large differential against a spot drag (Panel A, with the total return gap insignificant in Panel B). Inside signal months the composition matters. Downcurve signal months earn +5.5%/yr in total (spot −1.0 against a +6.5 differential), while all other signal months average −11.2 (spot −15.6 against +4.3). The wedge is +16.8%/yr (wild bootstrap $p = 0.05$ with signal runs as clusters). Two things are true at once in the downcurve months, and both matter for the compensation reading. The spot leg still depreciates—it is negative in 54% of downcurve signal months, against 41% outside the signal—so the regime does not switch the risk off; but the crash is a fraction of its size elsewhere (expected shortfall −3.7 against −9.0%/month, worst month −4.8 against −12.1), and the differential covers a much larger share of it. In the downcurve signal months in which the spot leg loses, the month's income covers 40% of the loss, against 14% in the other signal months and 17% outside the signal. Bearing the signal's risk is unpaid in a growth scare and paid in an inflation normalization—not because the currency cannot fall, but because the fall is small and the differential covers a good part of it—and the curve's shape says which is which in real time. The safe haven corroborates the typing. The yen appreciates in the non-downcurve signal months and depreciates in the downcurve ones (Section 4).

The portfolio implication is the carry overwrite named in Section 2.5. It stays invested when the cluster arrives in the downcurve configuration and exits (or goes anti-carry) only in the other shapes. The carry panel of Table 6 prices the modification. The overwrite-exit earns +5.6%/yr at a Sharpe ratio of 0.82 (against 0.79 unconditioned), and the overwrite anti-carry earns +7.0%/yr at 0.90, with the same halved worst month. The improvement is largest for anti-carry, which gains more from the shape consideration than the exit does (0.79 to 0.90 against 0.79 to 0.82), because the anti-carry position's mistake cost is double. A plain anti-carry position is short through exactly the months in which the carry portfolio is collecting its largest differential; removing them from the action set is what makes the aggressive option safe enough to prefer. Beyond the Sharpe ratios, the economic content is the two-sided decision of Section 2.5, the count for the risk side and the shape for the compensation side. Conflating them—exiting whenever risk is elevated, regardless of pay—leaves about a third of the mean improvement over the always-invested position on the table. The downcurve regime is the empirical implementation of that principle in this sample, not a separate discovery, and the principle, that a hedge exposes the investor to a state-dependent opportunity cost, is the more general of the two.

Three caveats apply. The downward-sloping evidence rests on two episodes—that is the nature of inflation eras—though the second (2023–24) mostly sits *outside* the signal state and still shows the compensation pattern (+3.5%/yr in non-signal downcurve months), the closest available out-of-sample check. The wedge is robust to how the regime is drawn (strict shape or tight curvature bands, wild p 0.02–0.05) but not to arbitrarily wide bands, and it is the *shape*, not the rate level, that prices it, since a high-rate dummy's wedge is half the size and insignificant ($p = 0.32$). And the downward-sloping dummy alone predicts nothing unconditionally ($p = 0.75$)—the compensation claim is conditional on the signal, which is the correct reading. The environment does not make carry better; it makes carry *paid for the risk the signal identifies.*

6 Systemic versus Idiosyncratic—and versus the U.S. Curve

The framework's identification claim is that the *cluster* carries the information. A single country's inversion is as likely idiosyncratic as systemic, and the cross-country configuration is the filter. (That bilateral exchange rates load heavily on common components is well documented, Verdelhan, 2018; the question here is whether the *curves'* common component is what predicts.) The claim has natural rivals at two levels, and this section confronts each at its own level. At the *portfolio* level—the carry and high-beta portfolios, spot and total—the rivals are the continuous global average slope, the U.S. curve, and the nine countries' own signals together. At the *bilateral* level—each currency against the dollar—the rival is the currency's own curve—its own inversion and its own slope information in the spirit of Chen and Tsang (2013).

A lone inversion is not a smaller version of the cluster; it is a different state, and the count itself shows it. Months with *exactly one* live inversion (88 of 458) are not weakly bad carry months—they are good ones. The carry portfolio earns +8.3%/yr in spot and +8.8 in total relative to the rest of the sample (wild $p = 0.005$ and 0.003; Table C.1), with a conditional mean total return of +11.3%/yr. A single inverted curve is typically a high-rate country's own cycle, which the diversified portfolio collects as interest rate differential rather than suffers as a crash. Months with exactly two live inversions are the crash state (−15.7 spot, −15.1 total, $p \leq 0.003$), and the sign flips at the second curve. The high-beta portfolio shows the same asymmetry (lone inversion +0.7, $p = 0.82$; two inversions −10.4, $p = 0.012$).

At the portfolio level, Table 7 runs the identification as a horse race and reports every rider. (A horse race and a control set are the same joint regression; the only difference is reporting—a control set reports one coefficient and treats the rest as nuisance, a horse race shows them all. No orthogonalization is imposed, because orthogonalizing would only reallocate the shared variance by an arbitrary ordering, whereas the joint regression lets every regressor take what it can, and the wild bootstrap partials the rivals under the null, so each bracketed p asks whether that state explains what the others do not.) Specification (1) is the cluster alone and reproduces Table 4. Specification (2) adds the *continuous global average slope*, its one-month change, and the *U.S. inversion*. All three are dead. The slope's coefficient is small and insignificant ($|t_{NW}| \leq 1.0$ in the carry columns), and the U.S. inversion is *positive* and insignificant (wild $p \geq 0.30$). The cluster is unchanged for carry (−13.5 spot, $p = 0.003$) and dips to marginal for the high-beta portfolio (−6.2, $p = 0.10$), the one place the slope level competes with the bare state (it returns under the risk controls of Section 10). Specification (3) adds the nine *own signals*, and the cluster *strengthens*—to −26.8 for carry spot and −12.0 for the high-beta portfolio, all four columns at $p \leq 0.040$—because the own signals absorb the benign single-curve months and leave the cluster identifying the synchronized ones. The own coefficients themselves are positive for eight or nine of the nine countries (several significantly so; individual coefficients in the replication log), the U.S. inversion stays dead, and the slope stays dead in the carry columns (its high-beta coefficient edges to the ten-percent boundary, the level effect the controls section prices).

Table 7: The portfolio horse race. Specification: $y_t = a + b\,\mathbf{1}\{\text{cluster}\}_{t-1} + \gamma' X_{t-1} + e_t$, with y the portfolio's annualized monthly return and X the rivals of the column, namely (1) none; (2) the continuous global average 10Y−2Y slope, its 1-month change, and the U.S. inversion (2s10s < 0, lagged, 57 months); (3) adding the nine countries' own signals (the paper's entry and exit applied to each single curve); all regressors lagged, 1988:01–2026:02. Each coefficient reports b, the Newey–West (12) t in parentheses, and—for binary states—the wild cluster bootstrap p on that state's episodes in brackets, other regressors partialled under the null. Specification (1), the cluster alone, is Table 4 and is omitted from the display.

	Carry spot		Carry total		High-beta spot		High-beta total	
	(2)	(3)	(2)	(3)	(2)	(3)	(2)	(3)
Cluster	−13.45***	−26.79***	−12.44**	−27.52***	−6.16	−11.99**	−5.66	−12.37**
	(−3.15)	(−4.88)	(−2.86)	(−4.92)	(−1.72)	(−2.02)	(−1.56)	(−2.07)
	[0.003]	[<0.001]	[0.012]	[<0.001]	[0.102]	[0.040]	[0.143]	[0.036]
Global slope	+0.21	+4.68	−0.64	+4.30	+2.40	+5.45	+2.40	+5.67
	(+0.08)	(+1.28)	(−0.22)	(+1.17)	(+0.90)	(+1.58)	(+0.88)	(+1.62)
Δ global slope (1m)	−14.97	−16.10	−15.14	−16.22	+11.87	+10.85	+11.41	+10.32
	(−1.15)	(−1.23)	(−1.17)	(−1.24)	(+0.92)	(+0.87)	(+0.89)	(+0.83)
U.S. inversion	+2.95	+2.23	+2.42	+1.63	+1.81	+1.27	+2.57	+2.00
	(+0.74)	(+0.51)	(+0.60)	(+0.37)	(+0.72)	(+0.38)	(+1.00)	(+0.57)
	[0.465]	[0.605]	[0.581]	[0.696]	[0.450]	[0.667]	[0.308]	[0.543]
Nine own signals (joint p)		[0.005]		[<0.001]		[0.009]		[0.004]

Each coefficient: b (%/yr for states, %/yr per pp for slopes), Newey–West (12) t in parentheses, and, for the binary states, wild cluster bootstrap p on that state's episodes in brackets, with the other regressors partialled under the null. Spec (2) adds the global slope, its 1-month change, and the U.S. inversion; spec (3) adds the nine own signals (coefficients in the replication log).
U.S. inversion alone (no cluster): carry spot −1.90 [0.484], total −1.34 [0.637]; high-beta spot −3.01 [0.205], total −2.03 [0.345].

Currency by currency against the dollar, the answer is the same, including against the currency's own curve. Table 8 moves to the bilateral level, where the currency's own curve is the natural rival; the dependent variable is the spot return, where the crash lives (Section 2.1). For every currency it reports the own signal alone, the cluster alone, both together, and the full specification—the cluster with the own signal, the currency's own-curve information à la Chen and Tsang (2013) (own slope, its one-month change, and the slope relative to the U.S. curve), and the global slope—with every coefficient carrying both the Newey–West t and the wild bootstrap p on its own state's episodes. Three facts. First, a country's own signal does not predict its own currency. Only the Australian dollar is marginally significant alone (−8.8, wild $p = 0.044$), and conditional on the cluster the own-signal coefficients are insignificant for all nine ($|t_{NW}| \leq 1.8$, wild $p \geq 0.22$). Second, the cluster's per-currency coefficients are essentially unchanged across the columns. The Australian and New Zealand dollars stay at -9 to -14%/yr (the wild p-values rise in the last column because seven regressors on fifteen episodes leave little bootstrap variation, and for the Australian dollar because its own signal overlaps the cluster in 42 of its 47 months), and the yen's point estimate, positive throughout, strengthens once its own slope is held fixed ($+2.4$ to $+6.4$), the safe-haven response. Third, the own-curve information is itself mostly insignificant. For each currency separately, a Wald test asks whether that currency's *own* curve variables—its own signal, its relative slope, and its relative level, jointly—add anything once the cluster is in the regression. The test is run nine times, once per currency, and rejects at the ten-percent level for exactly one of the nine, the yen ($p = 0.006$, driven by its relative slope), and for none of the other eight ($p \geq 0.11$). A currency's own curve, in other words, contains essentially no information for its own exchange rate beyond what the cross-country cluster already carries, so the Chen–Tsang variables and the cluster are not competing for the same variation. Individual-currency inference is weak by construction (one currency, fifteen episodes); it is the cross-sectional alignment of these coefficients with beta (Section 4) that carries the mechanism. The pooled panel with currency fixed effects puts the cluster at -4.7%/yr ($t_{NW} = -2.3$) with the own signals at zero (0.0, $t = 0.0$). The pooled bilateral coefficient is smaller than the portfolio coefficient of Table 4 by construction, since it is the average *level* response of nine currencies against the dollar—dollar movements included, long and short legs averaged rather than netted—while the portfolio number is the long-minus-short *spread*. The two measure different objects, and the spread is the one the carry trade holds.

The U.S. curve deserves its own verdict because it is the common practitioner proxy for "the" global signal. It does not predict the crash portfolio (-1.9%/yr on carry spot, wild $p = 0.48$, against the cluster's -12.9, $p = 0.001$; the high-beta portfolio gives the same verdict, $p = 0.21$ against 0.010), and the reason is coverage. The U.S. curve is inverted in 57 lagged months of the sample; the naive two-or-more count covers roughly three times as many (165 months) and the IYC signal 92, and the U.S. months are almost a strict subset—

Table 8: Each currency against the dollar. Specification: $\Delta s_{i,t} = a + b_{cl}\,\mathbf{1}\{\text{cluster}\}_{t-1} + b_{own}\,\mathbf{1}\{\text{own}\}_{i,t-1} + \gamma' Z_{i,t-1} + e_{i,t}$, spot returns (%/yr) on the currency's own signal alone (the paper's entry and exit applied to its own curve; ON = its months), the cluster alone, both together, and the full specification adding own-curve controls Z (own 10Y−2Y slope, its one-month change, the slope relative to the U.S.) and the global average slope (level and change), all lagged; 1988:01–2026:02. Each coefficient: b, Newey–West (12) t in parentheses, and wild cluster bootstrap p on that state's episodes in brackets, other regressors partialled under the null. The bracket in the last column reports, for that currency alone, the Newey–West Wald p of the null that its own signal, relative slope, and relative level are jointly zero given the cluster (one test per currency, not a test across currencies). Pooled row: currency fixed effects.

		Own signal alone		Cluster alone		Cluster + own signal				Full specification	
		own		cluster		cluster		own		cluster	
Currency	ON	b	(t_{NW})	b	(t_{NW})	b	(t_{NW})	b	(t_{NW})	b	(t_{NW})
AU	47	−8.76	(−1.65)	−13.21	(−1.83)	−14.18	(−1.38)	+2.19	(+0.23)	−12.92	(−1.31)
		[0.044]		[0.015]		[0.133]		[0.859]		[0.262, joint 0.408]	
CA	40	+0.71	(+0.19)	−4.61	(−1.33)	−6.01	(−1.45)	+4.50	(+1.04)	−5.55	(−1.46)
		[0.858]		[0.203]		[0.207]		[0.356]		[0.282, joint 0.801]	
CH	12	−20.12	(−1.91)	−0.89	(−0.16)	+1.28	(+0.21)	−20.95	(−1.81)	+1.15	(+0.18)
		[0.303]		[0.867]		[0.848]		[0.272]		[0.875, joint 0.115]	
EU	25	+5.26	(+0.62)	−1.19	(−0.20)	−3.40	(−0.51)	+8.13	(+0.80)	−4.00	(−0.61)
		[0.589]		[0.831]		[0.584]		[0.552]		[0.560, joint 0.488]	
GB	59	−3.84	(−0.59)	−3.00	(−0.63)	−2.00	(−0.45)	−2.82	(−0.44)	−0.44	(−0.10)
		[0.585]		[0.542]		[0.622]		[0.668]		[0.928, joint 0.486]	
JP	22	+4.58	(+1.03)	+2.35	(+0.58)	+1.68	(+0.38)	+3.41	(+0.71)	+6.37	(+1.32)
		[0.298]		[0.565]		[0.733]		[0.413]		[0.251, joint 0.006]	
NO	82	+4.53	(+0.84)	−3.96	(−0.71)	−6.85	(−1.15)	+7.43	(+1.42)	−8.17	(−1.23)
		[0.415]		[0.520]		[0.289]		[0.221]		[0.281, joint 0.233]	
NZ	48	−11.42	(−1.33)	−13.28	(−1.99)	−12.00	(−2.06)	−3.22	(−0.41)	−8.56	(−1.26)
		[0.282]		[0.011]		[0.011]		[0.748]		[0.220, joint 0.373]	
SE	13	−13.05	(−1.02)	−4.94	(−0.76)	−3.73	(−0.61)	−10.58	(−0.92)	−2.99	(−0.48)
		[0.497]		[0.416]		[0.526]		[0.540]		[0.662, joint 0.150]	
Pooled (FE)				−4.74	(−2.34)	−4.74	(−2.34)	−0.01	(−0.01)		

the U.S. proxy contributes only *one* month (2000:03) that neither measure covers, while missing two-thirds of the cluster's. The misses are not random. The Asian-crisis/LTCM period shows a U.S. inversion in one of the eighteen months in which two or more non-U.S. curves stood inverted (June 1998, between the paper's two 1997–98 episodes); the 1992 ERM collapse has none; and the U.S. 2s10s un-inverted in mid-2007, so the fifteen months leading into September 2008 carry no U.S. signal at all. The only episodes the U.S. proxy covers in full are the 2000 inversion and 2022–24; every episode that carries a carry disaster—1995, 1997–98, 2008, 2020—shows no U.S. signal at all. Global scares are visible in the cross-country configuration because they need not originate in, or even fully reach, the U.S. yield curve.

Table 9: EM spot returns on the IYC signal, per currency. Specification: $\Delta s_{i,t} = a_i + b_i\,\mathbf{1}\{\text{signal}\}_{t-1} + e_{i,t}$ (1996–2026; annualized; t_{ep} is the episode-clustered t on signal runs; wild bootstrap p).

Currency	b	t_{ep}	wild p	N	start
BRL	−9.37	−1.13	0.324	362	1996-01
CLP	−9.50	−1.32	0.219	362	1996-01
IDR	−31.02	−1.12	0.408	362	1996-01
INR	−7.98*	−2.14	0.061	362	1996-01
MXN	−11.44	−1.62	0.127	362	1996-01
KRW	−16.61	−1.57	0.153	362	1996-01
ZAR	−14.68**	−2.76	0.015	362	1996-01

7 Extension: Emerging Markets

Emerging-market currencies are the sharpest out-of-construction test available. They enter nowhere in the signal—no EM curve is counted, no EM rate is sorted—yet if the cluster marks a global growth scare, the currencies most levered to global conditions should depreciate on it. The test is not a mechanical extension. Uncovered interest parity behaves differently in emerging markets (Bansal and Dahlquist, 2000), so a developed-market bond state has no automatic claim on EM exchange rates.

The EM universe is the seven floating EM currencies with complete spot and money-market data, namely the Brazilian real, Chilean peso, Indonesian rupiah, Indian rupee, Mexican peso, Korean won, and South African rand. A completeness audit shows all seven are gap-free from January 1996 (the Chilean series is the binding start); the tables therefore use 1996 as the baseline start, with 2003—the point by which every crawling peg and band in the universe had been abandoned (the regime histories in Ilzetzki et al., 2019 date the last of them to the early 2000s)—as the regime-clean robustness sample. The results are stable across the two starts, so the choice is presentational, not substantive.

Three findings follow.

The export is directionally uniform and large (Tables 9–10). All seven currencies load negatively on the IYC signal, with a pooled effect of −14%/yr, roughly twice the G10 high-

Table 10: EM pooled panel. Specification: $y_{i,t} = a_i + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_{i,t}$, with y the spot change or the total return as labelled and the state the IYC signal, the naive synchronized-inversion count, or that count restricted to inversions at most six months old (currency fixed effects; t_{ep} clustered by state run; wild bootstrap on state runs).

Pooled specification	b	t_{ep}	wild p	N
Signal, spot	−14.37	−1.88	0.104	2534
Signal, total	−13.35*	−2.04	0.079	2533
Naive count, spot	−4.54	−1.08	0.359	2534
Naive count age ≤6, spot	−3.92	−0.52	0.766	2534

beta portfolio's effect. The same *construction* ordering as in the G10 sample (the disciplined signal against its cruder ancestors) also holds. The paper's signal dominates (−14.4 pooled spot), the naive inversion count is weak (−4.5), and its age-filtered version weaker still. The ordering replicates in currencies the signal has never seen.

Inference is marginal. The pooled wild bootstrap p is 0.079 in total returns and 0.104 in spot at the 1996 start (0.073 and 0.137 at the 2003 start); individually only the rand is significant (−14.7, $p = 0.015$), with the rupee marginal. The reason is structural. The signal's episodes in the EM sample amount to a dozen short runs concentrated in a handful of distinct crisis clusters, and run-clustered inference correctly treats them as few observations. The EM evidence is an economically consistent export with borderline standalone significance—corroboration of the mechanism, not an independent replication.

The downcurve regime travels to EM; the overwrite does not, for an economically coherent reason. The regime itself is global. When everything is constructed EM-natively (an EM inversion count over the seven EM curves, and the EM average curve's own shape, with the money-market rate as the short point), the EM downcurve months fall inside the same windows as the G10's—a brief mid-2007 window and 2023, the G10's defining 1988–91 era predating the EM data—and the same directional pattern appears. Relative to off-signal months, the EM own-signal costs the EM basket −10.6%/yr when the EM curve is not downward-sloping (run-clustered $t = -2.9$) but is roughly flat (+1.4 total) in EM-downcurve months; the within-signal wedge across shapes is +6.9%/yr ($t_{ep} = 2.1$ on just seven months—suggestive, not conclusive; Appendix I, Table I.1). What does *not* travel is the pay. The overwrite's logic requires a large interest rate differential in the regime, and the EM differential is small precisely then. Whether measured against the G10 regime (+2.9–3.2%/yr against +5.9–7.4 elsewhere) or the EM-native one (+3.4 against +4.5–4.7), the EM differential is compressed in the regime's defining condition—high U.S. rates. In the downcurve regime EM carry is *defanged but not paid*. The shape logic replicates, the compensation does not, as the interest-differential mechanism predicts. The overwrite is a statement about who is being paid, not a universal timing rule.

The G10 cluster beats the EMs' own curves. EM government-bond curves exist for all seven countries (with staggered starts), permitting the identification race within EM, which pits an EM own-inversion count (≥ 2 of the available EM curves inverted) against the IYC signal. The IYC signal carries nearly three times the loading (-13.6 against -5.2) and is essentially undented by including the EM count (the race's full output is in the replication package). Even for emerging markets, the informative curves are the developed markets', because the scare that crashes EM currencies is the global one, priced where curves are cleanest.

8 The Portfolio with Emerging Markets

The carry trade that practitioners actually run is not a developed-markets portfolio. The largest differentials are in emerging markets, and a rate-sorted portfolio drifts there mechanically. This section rebuilds the paper's two portfolios on the combined universe—the nine non-dollar G10 currencies plus the seven floating EM currencies of Section 7, sixteen in all, sorted top-3 minus bottom-3 exactly as in (2)—and asks how much of the G10 anatomy survives. The sample is 1996:01–2026:02, the period over which every EM spot and money-market series is complete. The sort concentrates where expected. Emerging markets fill 99% of the carry long-leg slots and 93% of the high-beta ones (in the carry sort the Brazilian real

Table 11: Conditional moments with and without emerging markets, 1996:01–2026:02. For each portfolio and leg, monthly returns off and on the (lagged) IYC signal, summarized by months, annualized mean (%/yr) and Sharpe ratio, skewness, and the monthly 5% value-at-risk, 5% expected shortfall, and worst month (%/month). Panel A: the paper's G10 portfolios on this subsample; Panel B: the same sorts on the sixteen-currency universe.

Series	State	N	Mean (%/yr)	Sharpe	Skew	$\text{VaR}_{5\%}$	$\text{ES}_{5\%}$	Worst
Panel A: the G10 portfolio								
Carry, spot	No signal	305	+2.96	0.44	−0.39	−3.31	−4.40	−6.04
	Signal on	57	−12.79	−1.24	−1.10	−6.80	−8.75	−12.09
Carry, total	No signal	305	+6.22	0.93	−0.38	−3.08	−4.12	−5.78
	Signal on	57	−8.67	−0.84	−1.09	−6.36	−8.41	−11.61
High-beta, spot	No signal	305	+2.03	0.26	−0.58	−3.95	−5.18	−9.19
	Signal on	57	−8.72	−1.10	−0.42	−4.70	−5.51	−6.85
High-beta, total	No signal	305	+4.07	0.53	−0.58	−3.78	−4.99	−8.97
	Signal on	57	−6.32	−0.79	−0.42	−4.41	−5.32	−6.69
Panel B: the G10+EM portfolio (16 currencies)								
Carry, spot	No signal	305	+0.46	0.04	−0.81	−5.23	−7.01	−19.25
	Signal on	57	−12.51	−0.63	−1.66	−10.77	−17.16	−26.18
Carry, total	No signal	305	+11.68	1.08	−0.38	−4.07	−5.73	−16.55
	Signal on	57	+2.51	0.12	−1.08	−9.35	−15.02	−23.43
High-beta, spot	No signal	305	−0.58	−0.06	−0.10	−4.81	−6.52	−9.77
	Signal on	57	−5.69	−0.39	−0.93	−6.76	−12.04	−14.88
High-beta, total	No signal	305	+6.79	0.66	+0.05	−4.09	−5.74	−8.92
	Signal on	57	+3.68	0.24	−0.48	−6.14	−10.64	−14.35

alone appears in 93% of months), and the funding leg is the familiar yen–franc–euro trio, so the G10+EM portfolio is, in practice, long EM against short DM funders.[1]

Table 11 gives the conditional distributions and Table 12 the state regressions, each cell carrying both the Newey–West t and the wild cluster bootstrap p on the state's episodes (the paper's convention, Section 3.4). Three facts organize the comparison.

First, the exchange-rate anatomy is the same. On the signal the G10+EM carry spot leg loses −12.5%/yr against the G10's −12.8, the loss again concentrates in the non-downcurve months (−14.7), and the downcurve stay-set is again flat. The point estimates are near-copies of the G10's; what changes is power. The EM legs' volatility is roughly twice the G10's, and

[1] A variant that restricts the universe to the three funders plus the seven EMs is qualitatively identical on every cell and is omitted; it is in the replication package (scripts 159–161), along with the within-EM long-short, which shows nothing on the signal—all seven EM currencies crash *together* on it, so a long-short inside EM nets out exactly the common component the signal identifies—and the EM-yield-curve version of the signal, which loses the race to the G10 cluster outright (its own coefficients are positive wherever the G10 signal's are negative).

twelve mostly short episodes cannot average it away, so the wild p-values sit at 0.18–0.24 rather than 0.003 and below. The spot side of the signal travels to the practitioner's portfolio directionally, not statistically.

Second, the interest rate differential changes the outcome entirely. The G10+EM differential averages +11.8%/yr—three and a half times the G10's +3.4—and it is *largest* in the growth-scare cell (+15.6%/yr in non-downcurve signal months, partly the defense hikes that crises themselves price). The consequence shows in the total-return rows of Table 11, where on the signal the G10 carry portfolio earns −8.7%/yr while the G10+EM portfolio earns +2.5—the differential absorbs, on average, the entire scare-month spot crash. The compensation question of Section 5.3 therefore has a different answer in EM. For the G10 portfolio the differential covers the crash only in the downcurve regime; for the EM-heavy portfolio it is a level, not a regime—large in every state—and the mean loss is paid for wherever it arrives.

Third, what the differential buys is the mean, not the tail, so the signal's value changes character rather than disappearing. The same signal months in which the G10+EM mean total is positive carry a 5% expected shortfall of −15.0%/month and a worst month of −23.4, roughly twice the G10 tail (−8.4 and −11.6), with skewness of −1.7 on the spot leg, so the signal marks the months in which the EM tail doubles. Figure 8 prices what that is worth. Exiting on the signal leaves the mean essentially unchanged (+9.8 against +10.2%/yr always-in—the forgone +15% differential and the avoided spot crash cancel on average) but removes a disproportionate share of the risk—the Sharpe ratio rises from 0.81 to 0.99, the expected shortfall shrinks from −7.9 to −5.4%/month, the worst month from −23.4 to −16.6, and the skewness from −0.98 to −0.25. For the G10 portfolio the signal times the *mean*; for the EM-heavy portfolio it trims the *tail* at zero mean cost. Two corollaries follow. Anti-carry is dead here (0.74), because being short a position that pays a +15% differential is the carry-premium risk of Section 2.5 at triple size. And the downcurve overwrite adds essentially nothing over the plain exit (1.00 against 0.99). With compensation high in every

Table 12: State regressions with and without emerging markets. Specification: $y_t = a + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_t$, y the portfolio's annualized monthly return, for the signal, its non-downcurve months, and its downcurve months; 1996:01–2026:02. Each cell: b (%/yr), Newey–West (12) t in parentheses, wild cluster bootstrap p on the state's episodes in brackets ($B = 9{,}999$); stars by the wild p.

	Signal		Excl. downcurve		Downcurve stay-set	
	b	(t_{NW})	b	(t_{NW})	b	(t_{NW})
Panel A: the G10 portfolio						
Carry, spot	−15.75***	(−3.02)	−17.59***	(−3.07)	+2.40	(+0.48)
	[<0.001]		[<0.001]		[0.710]	
Carry, total	−14.89***	(−2.83)	−16.68***	(−2.88)	+2.65	(+0.52)
	[0.003]		[0.003]		[0.677]	
High-beta, spot	−10.75***	(−3.94)	−12.41***	(−4.44)	+4.67	(+0.46)
	[0.001]		[<0.001]		[0.827]	
High-beta, total	−10.39***	(−3.77)	−12.13***	(−4.26)	+5.56	(+0.55)
	[0.001]		[<0.001]		[0.713]	
Panel B: the G10+EM portfolio						
Carry, spot	−12.97	(−1.75)	−14.68	(−1.86)	+3.46	(+0.46)
	[0.176]		[0.155]		[0.881]	
Carry, total	−9.18	(−1.34)	−10.28	(−1.39)	+1.65	(+0.23)
	[0.235]		[0.213]		[0.896]	
High-beta, spot	−5.10	(−0.93)	−6.15	(−1.02)	+4.17	(+1.57)
	[0.350]		[0.290]		[0.329]	
High-beta, total	−3.11	(−0.52)	−4.08	(−0.61)	+4.97	(+1.74)
	[0.659]		[0.595]		[0.392]	

state (against the funding currencies this portfolio borrows in, even where Section 7 finds the EM differential against the dollar compressed in the regime), there is no regime for the shape to price—the overwrite is a developed-markets phenomenon because only there is the compensation regime-dependent. This is the comparative static the two-sided decision of Section 2.5 predicts rather than a failure of the shape.

9 Application: The Stock Market

If the signal marks a global growth scare, its footprint should not be confined to currencies. Table 13 applies the identical state—built entirely from non-U.S. yield curves—to the S&P 500 over the same 1988–2026 sample.

The conditional anatomy replicates. Outside the signal, the index earns +10.4%/yr at a Sharpe ratio of 0.76; inside it, +1.9%/yr at 0.11, with skewness deteriorating from −0.5 to −1.1 and the expected shortfall from −8.5 to −12.7% per month; the probabil-

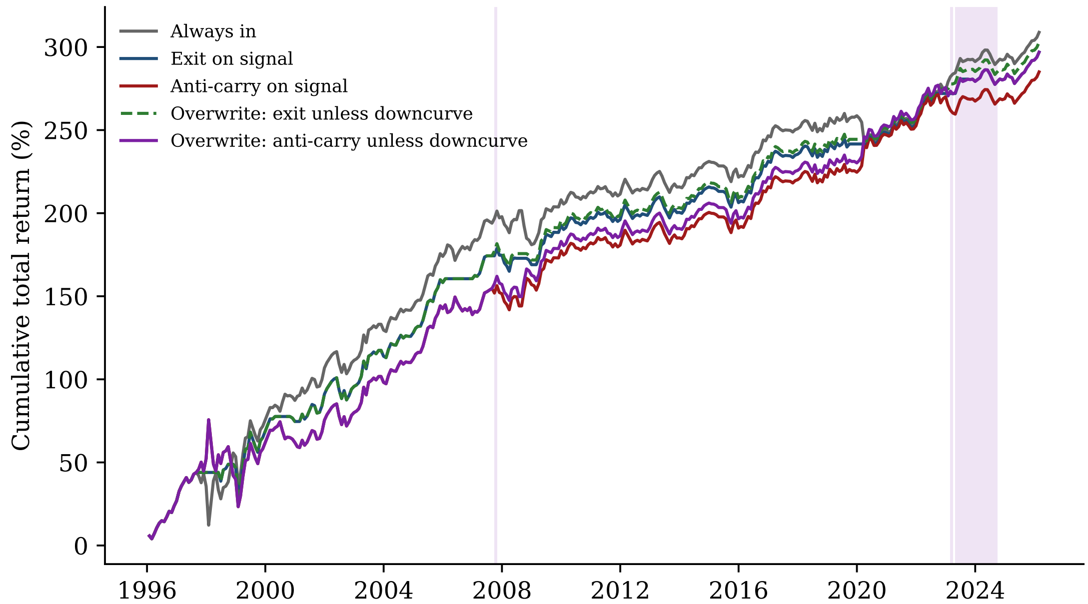

Figure 8: Cumulative total returns of the strategies on the G10+EM portfolio, 1996:01–2026:02; shaded bands mark downcurve months. The exit variants track the always-in line's mean while avoiding its deepest drawdowns (Sharpe 0.81 → 0.99; worst month −23.4 → −16.6); anti-carry does not pay, and the downcurve refinement adds nothing over the plain exit.

ity of a bottom-decile equity month roughly doubles (16.3% against 8.5%, difference wild $p = 0.087$).[2] And the compensation split carries over, with the damage concentrating in the non-downcurve signal months (−1.0%/yr, $\text{ES}_{5\%}$ −14.5), while the downcurve months—the inflation normalizations—are ordinary equity months (+6.7%/yr, normal tails). The same distinction that separates compensated from uncompensated carry separates benign from dangerous inversion months for equities. An inversion cluster in a high-inflation normalization is not an equity event; one in a growth scare is.

Two qualifications. First, episode-clustered inference is weaker here than for the currency portfolios. The mean effects (−8.5%/yr on the signal, −11.1 on non-downcurve months) carry wild p-values of 0.26–0.34, and the formal downcurve wedge—though its point estimate matches the moment split (+7.7%/yr)—is not significant for equities ($p = 0.49$); the

[2] The numerical coincidence with the high-beta portfolio's crash hazard (Section 5) is exact but mechanical. Any series' bottom-decile months number 46 on this 458-month calendar, and for both series 15 of the 46 fall inside the 92 signal months (15/92 = 16.3%, 31/366 = 8.5%)—the two sets of bottom-decile months themselves differ, and the two proportions are computed independently (replication scripts 129 and 134).

Table 13: The signal and the S&P 500, 1988:01–2026:02 (monthly total returns; $ES_{5\%}$ in %/month). Signal months are split by the downcurve regime as in Section 5.3. Panel B specification: $r_t^{SPX} = a + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_t$ (the wedge adds $\mathbf{1}\{\text{signal}\}_{t-1} \times \mathbf{1}\{\text{downcurve}\}_{t-1}$); wild bootstrap p on signal episodes.

	N	Mean (%/yr)	Sharpe	Skew	$ES_{5\%}$	Worst mo.
Panel A: conditional moments of monthly S&P 500 returns						
No signal	366	+10.42	0.76	−0.46	−8.45	−15.76
Signal on	92	+1.93	0.11	−1.10	−12.66	−18.56
Signal, non-downcurve months	57	−1.01	−0.05	−1.27	−14.53	−18.56
Signal, downcurve months	35	+6.71	0.46	−0.22	−8.52	−9.91
Panel B: state regressions and hazard (wild bootstrap p on signal episodes)						
S&P on signal: $b = -8.50$ ($t_{ep} = -1.17$, wild $p = 0.261$);						
on non-downcurve months: $b = -11.11$ ($p = 0.344$)						
bottom-decile hazard: 16.3% in-signal vs. 8.5% outside (difference wild $p = 0.087$)						
downcurve wedge: signal in non-downcurve months −11.43 ($p = 0.294$);						
signal × downcurve +7.71 ($p = 0.490$)						
Panel C: overlays (no transaction costs)						
Buy and hold	458	+8.72	0.60	−0.73	−9.91	−18.56
Exit on signal	458	+8.33	0.67	−0.36	−8.06	−15.76
Exit on non-downcurve months	458	+8.84	0.68	−0.38	−8.27	−15.76

compensation split is a currency-market result first, and for the S&P it is visible in the conditional moments rather than statistically separable. The reason is mechanical—equity volatility is twice the carry portfolio's and fifteen episodes are few. The equity application is a consistency check on the mechanism—the moments, the hazard, and the downcurve split all move the right way—not an independent significant result. Second, the overlays improve risk-adjusted rather than raw performance, because exiting on non-downcurve months lifts the Sharpe ratio from 0.60 to 0.68 and trims the tail while roughly preserving the mean. For a currency portfolio the signal is a first-order return event; for the equity index it is a risk event—which is what a growth-scare state priced first in bond markets should look like from the equity side.

10 More Than a Credit or Volatility Signal

Every result in the paper has been presented without risk controls, and deliberately so. The conditional facts are clearest in their raw form, and the natural objection—that a synchronized-inversion state is just the yield curve's way of spelling VIX, funding stress, or the credit cycle—deserves one dedicated confrontation rather than a column in every table. This section is that confrontation.

Table 14 runs the confrontation as a staircase, one panel per portfolio series, reporting every coefficient. The two gauges are the Chicago Fed NFCI—the broadest composite of funding, credit, and leverage conditions, available for the whole sample—and the VIX, each entering as a level and a one-month change (non-overlapping), all lagged one month. Specification (2) is the paper's main controlled test. Because the NFCI exists from 1971, it re-estimates every cell on the *full* 1988 sample with financial conditions held fixed. Specifications (3) and (4) add the VIX from its 1990 inception. The replication package confronts the signal with a wider battery of gauges (credit spreads, the trailing dollar return, global FX and Treasury volatility) on the shorter samples those series allow, with the same verdict.

The pattern is uniform. Nothing is absorbed, every coefficient holds or grows, and none loses significance. The carry spot effect is -12.8 with financial conditions held fixed on the full sample (wild $p < 0.001$) and -13.2 with all four controls ($p = 0.004$); carry totals hold as well, and the high-beta portfolio stays at -8.2 in spot and -8.0 in totals ($p \leq 0.025$). The controls themselves are worth reading. The one that matters is the *change* in financial conditions (ΔNFCI at t_{NW} -2.4 to -3.9; tightening conditions predict carry losses), while the VIX level enters *positively* (higher priced volatility predicts higher carry returns, the familiar volatility-premium pattern), and neither moves the signal. Conditioning on the market's standard stress gauges makes the signal look *stronger*, not weaker. Three observations explain why, and say what the state is.

First, *the state has no second-moment signature for the gauges to span.* Section 5 showed

Table 14: The signal against the risk gauges, one staircase of specifications per portfolio series. Specification: $y_t = a + b\,\mathbf{1}\{\text{signal}\}_{t-1} + \gamma' X_{t-1} + e_t$, with y the portfolio's annualized monthly return and X the controls of the column, namely (1) none; (2) NFCI level and one-month change—the NFCI exists from 1971, so this column covers the full 1988 sample; (3) VIX level and one-month change—*the VIX exists only from 1990 (the index's inception), so columns (3)–(4) begin 1990:12*; (4) all four. All regressors lagged one month; changes are one-month (non-overlapping). The signal reports b (%/yr), Newey–West (12) t in parentheses, and the wild cluster bootstrap p on signal episodes in brackets (controls partialled under the null); the controls, being continuous, report b and t_{NW} only.

	(1)	(2)	(3)	(4)
Panel A: carry, spot				
Signal	−12.85***	−12.84***	−15.28***	−13.18***
	(−3.16)	(−3.62)	(−3.00)	(−2.95)
	[0.001]	[<0.001]	[0.003]	[0.004]
NFCI (level)		+1.43		−6.04
		(+0.45)		(−1.58)
ΔNFCI (1m)		−48.32		−64.00
		(−3.78)		(−3.75)
VIX (level)			+0.26	+0.64
			(+1.17)	(+2.47)
ΔVIX (1m)			−0.33	+0.29
			(−0.96)	(+0.78)
N	458	458	423	423
Panel B: carry, total				
Signal	−11.31***	−11.56***	−14.62***	−12.55***
	(−2.66)	(−3.15)	(−2.86)	(−2.77)
	[0.003]	[<0.001]	[0.005]	[0.008]
NFCI (level)		+2.09		−5.97
		(+0.60)		(−1.60)
ΔNFCI (1m)		−48.17		−62.43
		(−3.86)		(−3.69)
VIX (level)			+0.25	+0.62
			(+1.13)	(+2.40)
ΔVIX (1m)			−0.32	+0.29
			(−0.92)	(+0.78)
N	458	458	423	423
Panel C: high-beta, spot				
Signal	−7.53***	−8.38**	−9.59**	−8.24**
	(−2.66)	(−2.91)	(−2.86)	(−2.49)
	[0.010]	[0.011]	[0.016]	[0.025]
NFCI (level)		+3.16		−3.73
		(+0.97)		(−1.04)
ΔNFCI (1m)		−29.99		−44.12
		(−2.44)		(−2.84)
VIX (level)			+0.42	+0.66
			(+1.91)	(+3.43)
ΔVIX (1m)			−0.31	+0.13
			(−0.97)	(+0.37)
N	458	458	423	423
Panel D: high-beta, total				
Signal	−6.82**	−7.86**	−9.22**	−7.95**
	(−2.36)	(−2.74)	(−2.78)	(−2.43)
	[0.013]	[0.013]	[0.016]	[0.025]
NFCI (level)		+3.66		−3.34
		(+1.06)		(−0.91)
ΔNFCI (1m)		−30.32		−43.67
		(−2.52)		(−2.82)
VIX (level)			+0.43	+0.65
			(+1.95)	(+3.46)
ΔVIX (1m)			−0.31	+0.14
			(−0.97)	(+0.37)
N	458	458	423	423

that for the high-beta portfolio the crash hazard doubles with essentially no change in variance, and for the carry portfolio the whole distribution shifts left (with only a statistically weak rise in variance). VIX and credit spreads are volatility and price-of-risk measures; a state that moves the first moment and the tail frequency, but not the variance, is orthogonal to them by construction. The signal is measuring something the gauges cannot—an expected-path event in bond markets, not a stress event in options or credit.

Second, *the comparison with cruder classifiers shows what absorption looks like when it does happen*. The naive inversion count's conditional mean—the state with no entry or exit discipline—is fully absorbed by the same gauges (under all four controls, carry $p \geq 0.58$, high-beta $p \geq 0.12$), and the absorption comes from the financial-conditions pair, not the volatility pair. The NFCI level and change alone take the naive count's carry-spot cell from $p = 0.047$ to $p = 0.15$ on the full sample, while the VIX pair leaves it significant ($p = 0.015$). A diluted state that mixes the crash season with long waiting periods co-moves with slow funding conditions and dies under them; the paper's signal, which isolates arrival and delivery, does not. Surviving controls is not a property of inversion information per se—it is a property of measuring that information correctly.

Third, *timing separates the signal from the gauges where it matters most*. The risk gauges are contemporaneous stress meters. VIX peaked with Lehman, not before it. The signal is an expectations aggregate, and it fired repeatedly through the fifteen months leading into September 2008—on in seven of them, including July through September 2008—while the U.S. curve was un-inverted and volatility was quiet. A practitioner conditioning on VIX or spreads learns that the crash is happening; one conditioning on the cluster learns that the scare has arrived while the forward is still priced for calm—which is precisely when the exit is cheap, and precisely the friction Section 2.3 names.

The currency literature's own factors do not span the state either. Regressing each portfolio's return on the lagged signal together with the *contemporaneous* dollar factor of Lustig et al. (2011), the average return of the nine currencies against the dollar, and the global FX volatility innovation of Menkhoff et al. (2012), measured as the change in the cross-sectional average absolute monthly spot return, asks whether the signal-month loss is merely factor exposure realized in those months. It is not. The carry spot coefficient is −11.5%/yr with both factors held fixed (wild $p = 0.002$; −12.0, $p = 0.004$ when the lagged NFCI and VIX controls are stacked on top), and the high-beta spot coefficient is −7.4 ($p = 0.009$; −8.7, $p = 0.011$ stacked). Substituting the innovation in G10 one-month implied volatility for the realized measure, on the shorter sample that series allows (1990 onward), leaves the carry cells significant ($p \leq 0.011$) while the high-beta cells keep their size but lose significance (−7.7 and −7.3, $p = 0.17$–0.18); computations in the replication package.

The synchronized-inversion state is an observable expected-growth state that the standard risk gauges neither contain nor explain.

11 Robustness

The full battery is in Appendix E; this section states what it contains and what it cannot fix. The naive inversion count fails the trader on both sides, forgoing more differential than it saves and switching off at the crash apex, which is why the latch matters and why the signal-gated strategy's worst month is half the baseline's; on the entry threshold, the baseline is not the extremum of the design grid—exactly two live inversions gives −15.7%/yr against the baseline's −12.9—and the paper reports the nested ≥ 2 rule because it is the one a trader can implement (Appendix C). No single episode carries the results (leave-one-episode-out worst case wild $p = 0.006$), and no headline cell is absorbed by the nested control sets (Section 10). The two halves of the sample each show the carry spot loss (−11.4%/yr before 2005, −15.9 after; Table E.1 in the appendix), with the high-beta loss a post-2005 result. The common term premium neither absorbs the headline as a control nor manufactures the inversions—83% survive stripping each curve's fitted exposure to the U.S. ACM slope premium, and the rule rebuilt on stripped slopes keeps the worst months, the high-beta loss, and the downcurve split (Appendix K). The downcurve wedge is robust to tight redefinitions of the regime, priced by the shape rather than the rate level, and conditional on the signal. Timing is executed rather than asserted, since a truncation audit rebuilds the signal and every conditioning series from data ending at each month-end and finds no classification or value ever changes when later data arrive, and an additional month of lag halves the carry coefficient and shrinks the high-beta one by a third, the forward-decay profile an ex-ante state variable should show.

What is weak, stated plainly. The signal has fifteen episodes; that is the inferential price of rare events, and the wild bootstrap on runs is priced for it. The downcurve-regime evidence rests on two inflation eras. The EM extension is corroborating rather than independently significant ($p = 0.079$–0.104 pooled at the 1996 start, 0.073–0.137 at the 2003 start), and the downcurve overwrite does not extend to EM at all (1.00 against 0.99)—the EM differential is a level, not a regime. The activity evidence rests on the leading indicator, since industrial production's decline does not survive lag controls. These are the boundaries of the claims, and the paper's language is calibrated to them.

12 Conclusion

When the G10 yield curves invert together, the world's bond markets are pricing a common expected slowdown—and the currencies that are long the global cycle, the carry trade's long leg by construction, subsequently crash. This paper builds the observation into a state variable with a trader's entry-and-exit discipline. It switches on when a fresh cluster forms and off only when two months of re-steepening confirm the easing has been delivered. The IYC signal's resulting 92 months contain the currency market's disaster risk—every conditional moment of the carry trade deteriorates inside them, the beta-sorted twin's crash probability doubles while its volatility stands still, and the worst months of four decades occur while the signal is on but after simpler inversion measures have quit. The signal survives the paper's full set of controls, is not the U.S. curve—which missed the Asian crisis and the road to Lehman—and reaches currencies that appear nowhere in its construction.

The paper's second theme is that risk management is a two-sided decision. Hedging removes the crash exposure but sacrifices the premium paid for bearing it, a carry-premium risk that an anti-carry position doubles, so the investor's problem is not simply whether to hedge when risk is high but whether the compensation is sufficient to justify remaining invested. The curve informs both sides, and practice tends to conflate them. The *count* of inversions measures the risk side, when disaster risk is elevated. The differential, observable in advance, and the *shape* of the inverted curves inform the compensation side, and the shape's reading rests on the sample's two inflation eras. The non-downcurve signal months are uncompensated (total returns of −11.2%/yr, −17.6 below all other months), while the downward-sloping high-rate configuration—short rates near nine and a half percent, every future rate expected lower—pays the carry position its largest interest rate differential through the inversion. The downcurve overwrite that stays invested in exactly those months, and steps aside in the others, is the empirical implementation of the principle in this sample, and the principle is the more general of the two; the emerging-market portfolio, in which the compensation is large in every state and the signal's value shifts from the mean to the tail, is its comparative static. Measuring risk tells you when to worry; measuring compensation tells you whether worrying is worth the differential you would give up.

The natural next steps are structural—embedding an observable disaster-probability state in a quantitative model of currency risk premia, and deriving the compensation regimes from a policy rule rather than documenting them from curve shapes. The empirical object this paper contributes is deliberately simple—one cluster, one latch, one shape, all publicly observable every month for forty years.

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A Data Appendix

A.1 Money-market rates

Table A.1 summarizes the one-month money-market rates used throughout, namely the nine non-dollar G10 currencies and the U.S. dollar (the funding numeraire and tenth rate). These are the rates that define the income leg in (1), the carry sort in (2), and the interest differentials in every table.

Table A.1: One-month money-market rates by country, 1988:01–2026:02 (percent per annum). Entries are the mean, standard deviation, minimum, maximum, and coverage. Seven of the ten series end in 2026:01, so 2026:02, the last portfolio-return month they position, is the sample end used throughout the paper; the AU, CA, and SE series continue beyond it and are truncated there.

Country	Mean	SD	Min	Max	First	Last
AU	5.08	3.58	0.03	18.00	1988-01	2026-02
CA	3.67	3.14	0.10	13.63	1988-01	2026-02
CH	1.56	2.70	−3.65	9.62	1988-01	2026-01
EU	3.00	2.85	−0.57	9.78	1988-01	2026-01
GB	4.37	3.88	0.04	15.25	1988-01	2026-01
JP	1.03	2.07	−0.08	8.62	1988-01	2026-01
NO	4.73	3.61	0.21	24.89	1988-01	2026-01
NZ	5.51	3.66	0.15	18.51	1988-01	2026-01
SE	3.83	4.02	−0.63	26.00	1988-01	2026-02
US	3.21	2.64	0.05	10.06	1988-01	2026-01

Figure A.1 plots the ten rates over time. Two features matter for the paper. First, the carry cross-section is persistent, not episodic. The Australian or New Zealand dollar holds one of the top two slots of the G10 rate ranking in 80% of all months (both sit in the top three in 64%), while the yen sits below 1% in every month from 1996 onward—the long and short legs of the carry trade are close to fixed identities. Second, the compensation environment moves with the eras of Table A.2. Rates start high and dispersed (G10 mean 8.8%, cross-sectional standard deviation 2.8pp in 1988–94), compress toward zero together over the 2010s (mean 0.9%, dispersion 1.2pp), and partially re-open after 2022. When the average level is high the spread the carry trade harvests is wide; the downcurve regime of

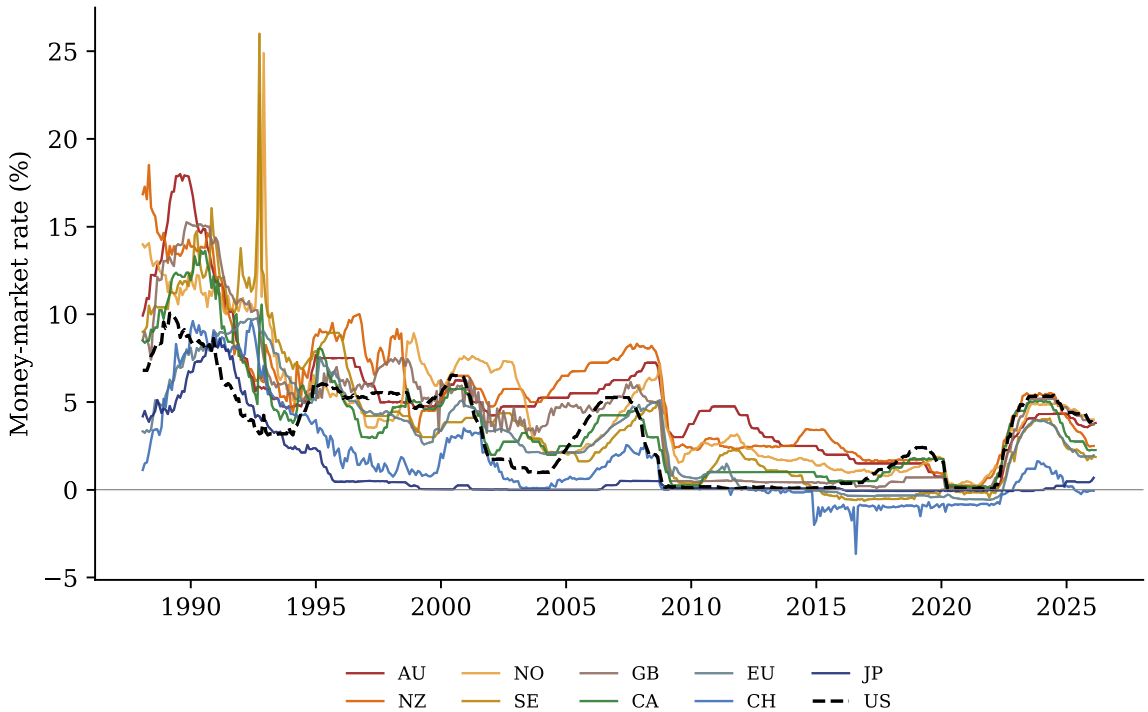

Figure A.1: One-month money-market rates, 1988–2026, for the nine G10 currencies (warm colors for the persistent high-rate bloc AU, NZ, NO, SE, GB; cool colors for EU, CH, JP; green for CA) and the U.S. (black, dashed).

Section 5.3 lives in exactly those years.

Figure A.2 plots the real-time expanding-window equity betas of (3), available for all nine currencies from 1989:04 (the estimation uses pre-sample data back to 1980 where available; the Australian dollar is the last currency to reach the 36-month minimum). The ranking is even more stable than the rate ranking—the smaller of the Australian and New Zealand betas exceeds the larger of the yen and franc betas in *every* month of the sample. This is the mechanism section's point in time-series form—the growth-exposure trait that the rate ranking prices is a persistent characteristic of each currency, not a rotating estimate.

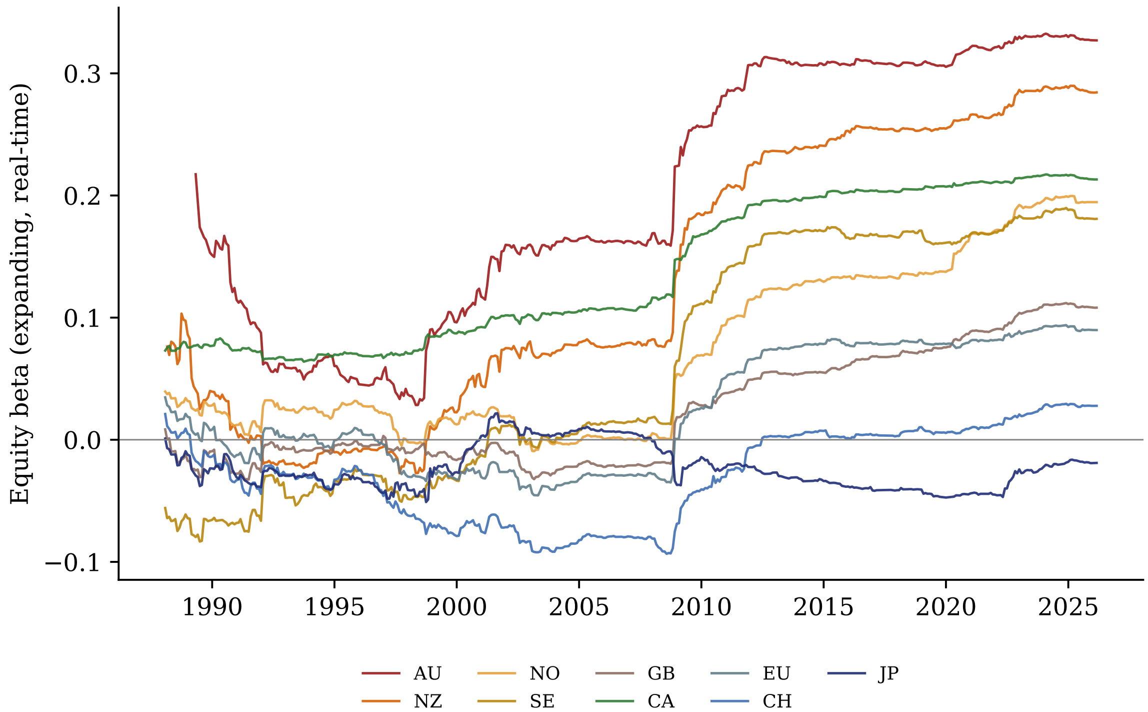

Figure A.2: Real-time equity betas, the expanding-window estimates $\hat{\beta}_{i,t}$ from (3), with the same color scheme as Figure A.1. The commodity currencies stay at the top of the ranking and the funding currencies at the bottom throughout the sample.

A.2 Eras and summary statistics

Table A.2 reports the era accounting. The sample opens in the post-inflation high-rate world (average G10 three-month yields of 8.5%), passes through the great moderation and the pre-crisis expansion, collapses into the Great Recession and a decade at the zero lower bound—where the carry trade's interest rate differential thins to +2.5%/yr and the signal falls nearly silent (three signal months in ten years)—touches the COVID trough, with negative average short rates, and ends in the post-COVID inflation era, where rates, inflation, the signal, and the downcurve regime all return. The two features of the table that the main text uses—the sixfold range of the interest rate differential across eras and the concentration of the downcurve months in the two high-nominal eras—are discussed in Section 3.

Table A.3 reports summary statistics for the nine currencies. The cross-sectional align-

Table A.2: The sample's economic regimes. For each era the table reports the number of months, the average G10 three-month yield (%), G10 CPI inflation (YoY %, nine currencies), the carry trade's income and total returns (annualized, %/yr), and the number of signal and downcurve months. Era boundaries are calendar conventions.

Era	Months	3M yield	CPI YoY	Income	Total	Signal	Downcurve
Post-inflation high rates (1988–94)	84	8.52	3.73	+5.94	+5.24	33	33
Great moderation / dot-com (1995–2000)	72	4.52	1.44	+4.86	+4.59	19	0
Pre-crisis expansion (2001–07)	84	3.41	1.66	+4.42	+9.56	17	1
Great Recession (2008–09)	24	2.66	2.03	+3.85	−1.36	6	0
Zero lower bound (2010–19)	120	0.80	1.39	+2.49	+1.70	3	0
COVID (2020–21)	24	−0.05	1.25	+0.91	−0.38	3	0
Post-COVID inflation (2022–26)	50	2.60	4.02	+3.11	+3.54	11	18

Table A.3: Summary statistics, 1988–2026. The table reports monthly spot returns vs. USD (annualized mean, volatility, skewness), the average money-market differential vs. the U.S., and the sample-average real-time equity beta.

Currency	Mean spot (%/yr)	Vol (ann.)	Skew	Avg. rate diff.	Equity beta
AU	−0.04	11.18	−0.46	+1.87	+0.20
CA	−0.14	7.49	−0.49	+0.46	+0.14
CH	+1.29	10.40	−0.08	−1.65	−0.03
EU	−0.26	9.69	−0.32	−0.21	+0.02
GB	−0.88	9.32	−0.61	+1.16	+0.02
JP	−0.67	10.51	+0.36	−2.18	−0.02
NO	−1.11	11.13	−0.33	+1.52	+0.07
NZ	−0.26	11.34	−0.33	+2.29	+0.13
SE	−1.16	11.06	−0.35	+0.62	+0.06

ment that Section 4 prices is visible in the raw statistics, with the persistent high-differential currencies (AU, NZ) at or near the top of the equity-beta ranking and the funding currencies (JP, CH) at the bottom.

A.3 Other series and coverage

Spot exchange rates are end-of-month, all nine currencies vs. USD, gap-free over 1988:01–2026:02 (the euro is spliced from the Deutsche mark before 1999). *Government yields* at 3-month, 2-year, and 10-year tenors are complete over the sample for all nine countries plus the U.S.; the 10-year/2-year pair defines inversion and the three tenors define the average curve's shape. The *U.S. equity market return* is the monthly total return, used in the beta estimation (3) and as the equity outcome of Section 9. Among the *controls*, the NFCI is available for the whole sample (the series begins in 1971) and the VIX begins in 1990 (the index's inception—our series starts 1990:10), so the controlled samples' start dates (the full sample for the NFCI specifications, 1990:12 for those with the VIX) are bound by data inception, not chosen. *CPI* is the equal-weighted average of the nine countries' year-on-year CPI inflation, complete over 1988:01–2026:02, from Global Financial Data. Seven countries publish a monthly index; Australia and New Zealand publish the CPI quarterly, so their year-on-year inflation is computed on the quarterly index, dated to the quarter's last month, and carried forward over the next two months, so that each month uses the latest published quarter. The average is lagged two months wherever it conditions a state. It is used only for the era and regime characterizations. *Activity* is a proxy. Industrial production (industry excluding construction, calendar and seasonally adjusted, OECD) is monthly for Canada, Germany, the United Kingdom, Japan, Norway, and Sweden from 1985 and for Switzerland from 2010:10; Australia and New Zealand publish it quarterly and are excluded. The G10 series is the equal-weighted average of the monthly log changes of the countries available in each month, cumulated, so Switzerland's late entry creates no level jump. The OECD composite leading indicator is available for Australia, Canada, Germany, Japan, and the United Kingdom, aggregated the same way. A country-level panel with country fixed effects reproduces the aggregate projections (replication package, script 178). Emerging-market data are described in Section 7 and Appendix I.

B Derivations

Proposition 1 (frictionless benchmark). From $rx_{t+1} = f_t - JZ_{t+1} + \eta_{t+1}$ and the pricing condition $\mathbb{E}_t[M\, rx_{t+1}] = 0$ we obtain $f_t\, \mathbb{E}_t[M] = J\, \mathbb{E}_t[MZ_{t+1}]$. Since $\mathbb{E}_t[MZ_{t+1}] = \mathbb{E}_t[M \mid$

$Z_{t+1} = 1]\,\lambda_t^P$ and $\mathbb{E}_t[MZ_{t+1}] = \mathrm{Cov}_t(M, Z_{t+1}) + \mathbb{E}_t[M]\,\lambda_t^P$,

$$\kappa_t^{\mathrm{fric}} = \frac{f_t}{\lambda_t^P J} = \frac{\mathbb{E}_t[M \mid \mathrm{crash}]}{\mathbb{E}_t[M]} = 1 + \frac{\mathrm{Cov}_t(M, Z_{t+1})}{\lambda_t^P\,\mathbb{E}_t[M]} \;\geq\; 1,$$

with strict inequality when $\mathrm{Cov}_t(M, Z_{t+1}) > 0$. With CRRA utility and a recession consumption contraction b, $\mathbb{E}[M \mid \mathrm{crash}]/\mathbb{E}[M] \approx (1-b)^{-\gamma} > 1$.

Proposition 2 (conditional loss). The predetermined forward is $f_t = c_t + \hat{\Lambda}\lambda^R\,\mathbb{E}_{t-1}[p_t]\,J$ (10) with $\mathbb{E}_{t-1}[p_t] \approx q_0$ at a fresh onset from the anchored prior. The true conditional hazard at the onset is $\lambda_t = \lambda^R P^\star$. Hence, with $c^\star \equiv \lambda^R J(P^\star - \hat{\Lambda}q_0)$,

$$\kappa_t = \frac{f_t}{\lambda^R P^\star J} = \frac{c_t + \hat{\Lambda}\lambda^R q_0 J}{\lambda^R P^\star J} \;<\; 1 \iff c_t < c^\star,$$

and the predictable excess return is $\mathbb{E}_t[rx_{t+1}] = f_t - \lambda_t J = c_t - c^\star$, negative in the same event; for $\hat{\Lambda}$ near one, $c^\star \approx J\lambda^R\,\Delta p_t$, the un-priced belief innovation.

Lemma 1 (Breadth sufficiency)**.** *Under Assumption 2 the mean slope is exactly sufficient for g_t; coarsened to inversion indicators, the count k_t is sufficient within the binary observation set.*

Proof of Lemma 1, and the calibrated threshold. With $S_i \mid g$ i.i.d. normal and inversion probabilities $q_s = \Phi(-\mu_s/\sigma)$, the likelihood of the binary vector depends on the data only through k, because $\Pr(\{I_i\} \mid s) = q_s^k(1-q_s)^{n-k}$; this is the sufficiency claim. The log posterior odds are

$$L(k) = \mathrm{logit}(q_0) + n\log\frac{1-q_R}{1-q_N} + k\,\ell, \qquad \ell = \log\frac{q_R(1-q_N)}{q_N(1-q_R)} > 0,$$

so $\Pr(R \mid k)$ is logistic and increasing in k. False synchronization is quadratically rare. For small q_N, $\Pr(k \geq 1 \mid N) \approx nq_N$ while $\Pr(k \geq 2 \mid N) \approx \binom{n}{2}q_N^2 \approx \frac{1}{2}(nq_N)^2$.

Proposition 5 (A calibrated threshold)**.** *If the action threshold is the crash break-even* $p^\star = f/(\lambda^R J)$ *rather than one half, then*

$$k^\star = \lceil (\mathrm{logit}(p^\star) - L(0))/\ell \rceil .$$

At the calibration $q_N = 0.05$*,* $q_R = 0.60$*,* $q_0 = 0.10$*,* $n = 9$*,* $p^\star = 0.03$*, the posterior at* $k = 0, 1, 2, 3$ *is* 0.00005*,* 0.001*,* 0.04*, and* 0.51*, so* $k^\star = 2$*.*

Status: calibrated. Under other parameters $k^\star$ rounds to three; the paper does not lean on this crossing—the main-text threshold rests on the $O((nq_N)^2)$ argument and on the empirical grid of Appendix C.

Proposition 3 (conditional Fama slopes). Let currency i's appreciation be $\Delta s_{i,t+1} = -J_i Z_{t+1} + \eta_{i,t+1}$ with crash exposure aligned to the differential, $J_i = \varphi f_i$, $\varphi > 0$. *Off signal.* Here $\lambda^P \approx 0$, so $\mathbb{E}[\Delta s_i] \approx 0$ for all i and the population regression of Δs on f has slope ≈ 0 (any positive slope in data reflects sampling and omitted appreciation trends); the puzzle relative to the UIP benchmark of -1 is immediate. *On signal.* The conditional hazard is $\lambda^R P^\star$, so $\mathbb{E}[\Delta s_i] = -\lambda^R P^\star J_i = -\lambda^R P^\star \varphi f_i$. The cross-sectional slope on f_i is $-\lambda^R P^\star \varphi$. Using the basket identity $\kappa = f/(\lambda^R P^\star J) = 1/(\lambda^R P^\star \varphi)$ (with $J = \varphi f$ evaluated at the basket), the slope is $-1/\kappa$. Fair pricing ($\kappa = 1$) restores UIP's -1; the conditional loss ($\kappa < 1$) pushes the slope below -1.

Proposition 4 (exit rule). $\mathbb{E}_t[rx_{t+1}] = f_t - \lambda^R P^\star J \lesseqgtr 0$ as $\kappa_t = f_t/(\lambda^R P^\star J) \lesseqgtr 1$; by Proposition 2 this is $c_t \lesseqgtr c^\star$. Unwinding to cash yields zero; going anti-carry yields $-\mathbb{E}_t[rx_{t+1}]$. Hence, for a risk-neutral one-period position, hold iff $\kappa_t \geq 1$, and the optimal use of a regime variable that indexes whether the interest rate differential covers the expected loss (the curve's shape) is exactly the overwrite. With risk aversion the hold region shrinks (staying adds variance), so $\kappa_t \geq 1$ is the outer bound of the hold region, not its exact boundary.

C Signal-Construction Robustness

The signal has two constructed elements, the cluster size at entry and the confirmation length at exit. Both are set on simple grounds in the text; this appendix tabulates the alternatives.

Table C.1: Cluster-size grid: the state defined by the number of live inversions—at least n ($n = 1, 2, 3, 4$) and exactly n ($n = 1, 2$)—with the identical per-curve entry and exit rules. State regressions $y_t = a + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_t$ as in Table 4 (bare, 1988:01–2026:02; wild bootstrap p on the respective state's episodes).

Live inversions	ON mo.	Carry spot		Carry total		High-beta spot		High-beta total	
		b	wild p	b	wild p	b	wild p	b	wild p
≥ 1 (any inversion)	180	−3.23	0.230	−1.90	0.513	−4.62***	0.008	−4.16**	0.014
$= 1$ (exactly one)	88	+8.33***	0.005	+8.78***	0.003	+0.69	0.818	+0.66	0.831
≥ 2 (baseline)	92	−12.85***	0.001	−11.31***	0.003	−7.53**	0.010	−6.82**	0.013
$= 2$ (exactly two)	48	−15.72***	0.001	−15.13***	0.002	−10.43**	0.012	−10.31**	0.011
≥ 3	44	−6.77	0.405	−4.57	0.694	−2.64	0.478	−1.47	0.711
≥ 4	22	−0.75	0.879	+2.10	0.672	+0.26	0.967	+1.87	0.736

Table C.2: Exit-confirmation grid: the signal rebuilt with the exit requiring K consecutive re-steepening months ($K = 1, 2, 3$), identical entry rule. Specification as in Table C.1.

Exit rule	ON mo.	Carry spot		Carry total		High-beta spot		High-beta total	
		b	wild p	b	wild p	b	wild p	b	wild p
1 re-steepening month	39	−12.14*	0.059	−10.49	0.112	−10.35**	0.035	−9.39*	0.051
2 consecutive (baseline)	92	−12.85***	0.001	−11.31***	0.003	−7.53**	0.010	−6.82**	0.013
3 consecutive	191	−3.59	0.161	−2.11	0.491	−4.23***	0.001	−3.58**	0.017

Table C.3: Slope-definition robustness: the signal rebuilt on 10Y−3M slopes instead of 10Y−2Y, identical entry and exit rules. Specification as in Table C.1.

Slope definition	ON mo.	Carry spot		Carry total		High-beta spot		High-beta total	
		b	wild p	b	wild p	b	wild p	b	wild p
10Y−2Y (baseline)	92	−12.85***	0.001	−11.31***	0.003	−7.53**	0.010	−6.82**	0.013
10Y−3M	107	−4.68	0.146	−4.02	0.260	−5.25**	0.011	−5.00**	0.034
10Y−2Y, excl. downcurve	57	−18.07***	0.001	−17.61***	0.002	−12.89***	<0.001	−12.59***	0.001
10Y−3M, excl. downcurve	74	−5.73	0.177	−6.10	0.201	−7.56***	0.003	−7.94***	0.004

Overlap of the two signals: 58 months. The 10Y−3M signal's downcurve stay-set (33 months) is flat (carry total +1.59, $p = 0.371$): the downcurve filter behaves identically on either construction.

A third construction check replaces the 2-year tenor with the 3-month bill (Table C.3). The result is qualitatively intact but quantitatively weaker, and the asymmetry is informative. Every sign is preserved and the high-beta portfolio remains significant (−5.3 spot, $p = 0.011$; −5.0 total, $p = 0.034$), but the carry-portfolio cells lose significance. The 10Y−3M signal is also a substantially different classifier (107 months, only 58 shared with the baseline). The 3-month end is dominated by the realized policy rate, so it inverts later in hiking cycles and misses the arrival phase that carries the loss. That the choice of tenor pair changes what a spread predicts is a general point (Engstrom and Sharpe, 2019); here the 2-year belly—the market's expected-path tenor—is load-bearing for the carry result, consistent with the expectations-hypothesis reading in Section 2.2.

Both grids say the same thing, namely that the baseline sits where the signal works and the alternatives fail in the directions the text's arguments predict. On the cluster size (Table C.1), accepting *any single inversion* dilutes the state to 180 months and washes out the carry effect (-3.2%/yr, $p = 0.23$), and the exact-count rows say why. Months with *exactly one* live inversion are positive carry months (+8.3 spot, +8.8 total, $p \leq 0.005$), so the "any inversion" state averages a benign idiosyncratic state with the crash state—idiosyncratic inversions are not noise around the systemic signal but a different state with the opposite sign (Section 6). Exactly two live inversions is the sharpest cell (-15.7 spot, -15.1 total); requiring three arrives too late and holds too few months (44), leaving negative but powerless point estimates, and requiring four holds so few (22) that the point estimates flip sign. The baseline is therefore not the maximum of the design space—the exact-two threshold produces the larger coefficient—and the paper reports the nested ≥ 2 rule because it is the one a trader can implement, since an exact-count state switches off the month a third curve inverts, which no manager watching a scare spread would do. The timing evidence says the same thing from the other side. Months with three or more live inversions sit deeper into an episode (median episode age six months against three, with eleven of the fifteen episodes starting at exactly two), and within the signal a three-or-more dummy adds nothing (wild $p \geq 0.27$; replication package), so the count beyond two is late-episode breadth, not additional risk. On the exit (Table C.2), a single re-steepening month exits too fast—the state shrinks to 39 months, stays large on spot (-12.1) but turns borderline there ($p = 0.06$) and fragile in totals ($p = 0.11$); three consecutive months waits through too much of the recovery, swelling the state to 191 months and diluting it toward the naive count (-3.6, $p = 0.16$). One is too noisy an exit, three too slow. The results concentrate at the baseline and degrade symmetrically away from it, as the text asserts; nothing here was tuned beyond that.

D The Naive-Count Benchmark

The natural benchmark for the paper's signal is the naive inversion count—at least two of the nine curves inverted, with no entry or exit discipline. Section 3 introduces it as the baseline and the two failures the signal's rules repair; this appendix contains the full anatomy.

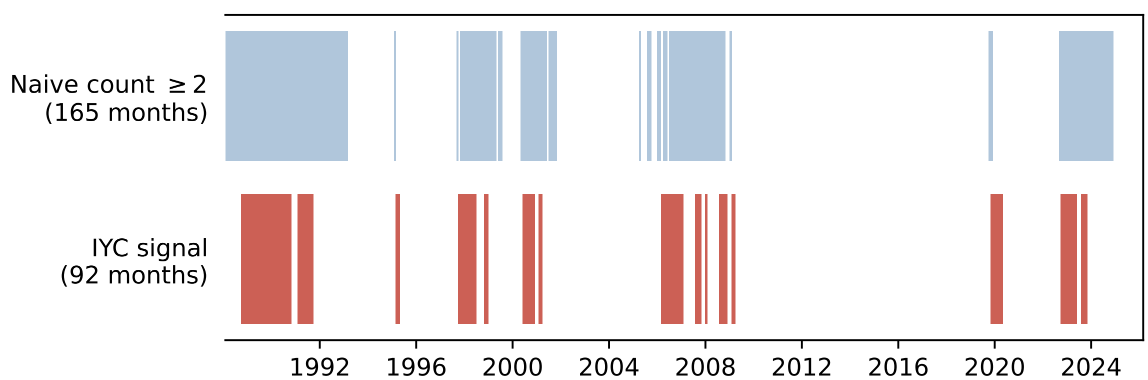

Figure D.1: How long each classifier is on. The naive count ($\geq$ 2 of 9 inverted, 165 months, top) is shown against this paper's signal (92 months, bottom), 1988–2026. The raw count stays on through long stretches in which nothing happens—most visibly 1988–93, 2006, and 2022–24—while the signal's episodes are shorter and end with the re-steepening confirmation.

Figure D.1 shows the basic problem. The naive count is on 36% of the sample—nearly twice the signal's 20%—because once a cluster forms it tends to stand for years. Table D.1 quantifies what that time-on costs. For each classifier it reports how many of the carry trade's worst-5% and worst-10% months fall inside the state, the *concentration* ratio (share of tail months captured divided by share of calendar occupied—1.0 is chance), and the tail rate inside the state. The naive count captures the most tail months in absolute terms (9 of 23) but only by being on the longest, so that per month of exposure its concentration is

1.09×—indistinguishable from chance, with an in-state tail rate of 5.5% against a 5.0% base; filtering the count by the downcurve regime lifts it only to 1.23×. The signal concentrates the same tail risk at 1.73×, and the downcurve-filtered signal at 2.45×, with one in four of its months a worst-decile month. The contrast sharpens at the extreme tail. The five worst months of the sample—the bottom one percent, namely October 2008, March 1995, March 2020, September 2008, and October 1998—all fall inside the signal's state (a concentration of 5.0×, rising to 8.0× for the filtered signal), while the naive count holds only two of them, its lagged state having switched off before each of the three worst. The naive count is not wrong about where the risk is; it is simply on for so long that being inside its state carries almost no information per month.

Table D.1: Tail capture per classifier. The table reports how much of the carry trade's worst months each state contains, against how much calendar it occupies. Concentration = capture share ÷ time share; tail months are the full-sample worst 1%, 5%, and 10% of carry total returns, 1988:01–2026:02; all states lagged one month.

Classifier	Months on	Time share	Tail captured	Concentration	In-state rate
Worst 1% of carry-total months (5 months ≤ −6.03%/month)					
Naive count ≥ 2	165	36%	2/5 (40%)	1.11×	1.2%
Naive count excl. downcurve	113	25%	2/5 (40%)	1.62×	1.8%
IYC signal	92	20%	5/5 (100%)	4.98×	5.4%
IYC signal excl. downcurve	57	12%	5/5 (100%)	8.04×	8.8%
Worst 5% of carry-total months (23 months ≤ −3.75%/month)					
Naive count ≥ 2	165	36%	9/23 (39%)	1.09×	5.5%
Naive count excl. downcurve	113	25%	7/23 (30%)	1.23×	6.2%
IYC signal	92	20%	8/23 (35%)	1.73×	8.7%
IYC signal excl. downcurve	57	12%	7/23 (30%)	2.45×	12.3%
Worst 10% of carry-total months (46 months ≤ −2.32%/month)					
Naive count ≥ 2	165	36%	19/46 (41%)	1.15×	11.5%
Naive count excl. downcurve	113	25%	16/46 (35%)	1.41×	14.2%
IYC signal	92	20%	15/46 (33%)	1.62×	16.3%
IYC signal excl. downcurve	57	12%	14/46 (30%)	2.45×	24.6%

The consequences run through the paper's numbers. As an overlay, exiting carry on the naive count *loses* 1.3%/yr against the always-invested baseline. The months it exits average +3.5%/yr in total returns, so the forgone interest rate differential dwarfs the crashes avoided. As a risk classifier it fails at the worst possible moments. The carry trade's three worst months in the sample (October 2008, March 1995, March 2020) all occur with the lagged naive count at *one*—below its own threshold—because the delivered easing that constitutes the crash re-steepens curves mechanically; only the signal's confirmation latch stays on through delivery. Its conditional mean is fully absorbed by the paper's control set (carry $p \geq 0.58$, high-beta $p \geq 0.12$), and the absorption comes from the financial-conditions pair rather than the volatility pair (Section 10)—the behavior of a diluted state that mixes the crash season with years of waiting.

Two natural repairs suggest themselves, and Table D.2 runs both. The first weights the count by the reciprocal of its age, so that fresh cluster months count fully and stale ones fade. This works, which shows that dilution is the mechanism—the carry spot effect revives from -3.2 to -10.4%/yr, totals from -1.1 to -9.8, and the high-beta portfolio from -4.7 to -14.9. The second repair is the paper's own downcurve filter, which drops the count's downcurve months on the logic that the compensated regime is where the count wastes most of its time (52 of its 165 months). For the carry portfolio this fails outright—the filtered count is -3.3 in spot ($p = 0.24$) and -1.7 in totals ($p = 0.58$), and its tail concentration rises only to 1.23×/1.41× (Table D.1) against the signal's 1.73×/1.62×. The downcurve months were not where the carry dilution binds, and the filtered count inherits the count's fatal timing. The three worst carry months all remain outside it, because the count itself is off at those dates. The overwrite wedge is likewise a signal-months phenomenon, not a general property of inversion months, as the downcurve gap in carry totals within the naive count's months is $+2.0$ ($t_{NW} = 0.6$), against $+16.8$ ($t_{NW} = 3.2$) within the signal's. Both repairs share a ceiling. Any transformation of the naive count is bounded by the count's month-set, which contains only nine of the twenty-three worst months at any choice of weights, and under the paper's control set the filtered carry repair dies ($p = 0.27$) while the age-weighted one fades to marginal (-7.6, $p = 0.08$, against the signal's -13.2 at $p = 0.004$). One asymmetry deserves note. For the high-beta portfolio—whose loss is an arrival phenomenon—both repairs work (the age-weighted count is the strongest conditioning of all, -14.9 bare and -13.8 under controls; the filtered count sharpens to −6.4 spot and −6.1 total), because arrival months sit inside the count's own state. For the carry trade, whose losses require staying on through delivery, only the latch reaches them. Dilution is fixable by weights; misclassification is not. The signal's entry and exit rules are therefore not refinements of the naive count; they are the difference between a state that contains the risk and one that mostly contains waiting.

Table D.2: Repairing the naive count. State regressions $y_t = a + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_t$ as in Table 4, with the conditioning variable the naive count, the count restricted to non-downcurve months, the count weighted by 1/age of the standing cluster, and the paper's signal (with and without the same filter); bare, 1988:01–2026:02. Wild bootstrap p on the respective state's episodes.

Conditioning	Carry spot		Carry total		High-beta spot		High-beta total	
	b	wild p	b	wild p	b	wild p	b	wild p
Naive count ≥ 2	-3.18^{**}	0.047	-1.06	0.670	-4.73^{***}	0.005	-4.06^{**}	0.012
Naive count, excl. downcurve	-3.31	0.238	-1.73	0.581	-6.43^{***}	0.003	-6.13^{***}	0.006
Naive count, 1/age-weighted	-10.43^{**}	0.040	-9.75^{*}	0.057	-14.87^{***}	0.001	-14.46^{***}	0.002
The signal	-12.85^{***}	0.001	-11.31^{***}	0.003	-7.53^{**}	0.010	-6.82^{**}	0.013
The signal, excl. downcurve	-18.07^{***}	0.001	-17.61^{***}	0.002	-12.89^{***}	<0.001	-12.59^{***}	0.001

Under the paper's control set (NFCI level and 1m change, VIX level and 1m change; 1990:11–):
naive count carry spot −1.91 ($p = 0.576$), total −0.63 ($p = 0.847$);
1/age carry spot −7.55 ($p = 0.084$), high-beta spot −13.82 ($p = 0.008$);
excl.-downcurve count: carry spot −3.99 ($p = 0.269$), high-beta spot −7.04 ($p = 0.035$).

E Robustness Details

The program is organized around the paper's load-bearing choices, namely the signal's entry and exit rules, the leave-one-episode-out stability of the results, the controls, the downcurve regime's definition, and timing. "Survives" always means the wild cluster bootstrap on signal runs.

Why the latch, and not a simpler measure. The naive inversion count fails the trader on both sides—as an overlay it forgoes more of the interest rate differential than it saves, and as a classifier it switches off at the crash apex (the carry trade's three worst months all occur with the lagged count at *one*, below its own threshold, because delivered easing re-steepens curves mechanically). The confirmation latch is what keeps the classifier on through delivery, and it is why the signal-gated strategy's worst month is half the baseline's. On the entry threshold, the baseline is not the extremum of the design grid—exactly two live inversions gives -15.7%/yr against the baseline's -12.9—and the paper reports the nested ≥ 2 rule because it is the one a trader can implement (Appendix C states the comparison). Appendix D collects the full benchmark comparison, including a visual comparison of how long each classifier is on; Appendix C varies the cluster size ($n = 1$ to ≥ 4) and the confirmation length ($K = 1, 2, 3$).

No single episode carries the results. Leaving out each signal episode in turn, the high-beta portfolio's signal exposure survives every drop (worst-case wild $p = 0.006$), and the carry-portfolio coefficients stay within a third of their full-sample values. The functional form of episode freshness (reciprocal age, square root, exponential decay, plain dummy) is immaterial.

Controls. Section 10 is the dedicated treatment, and it finds that no headline cell is absorbed or loses significance under the nested control sets.

The two halves of the sample. The rule has no fitted parameters, only three design choices whose rationales precede the data (Section 2), so the honest version of an out-of-sample check is stability across halves of the sample rather than a held-out period, since the choices were made knowing that 2008 and 2020 happened. Table E.1 re-estimates the headline on 1988:01–2004:12 and 2005:01–2026:02. The carry spot loss is there in both halves (-11.4%/yr, wild $p = 0.037$, and -15.9, $p = 0.010$), the later half, which contains 2008 and 2020, is the stronger one, and the high-beta portfolio's loss is entirely a post-2005 result (-11.9 spot, $p = 0.010$, against -4.0 and insignificant before). The three worst carry months of each half sit inside the state in the later half (2008-09, 2008-10, 2020-03) and two of three in the earlier one (1995-03 and 1998-10; November 1992, the ERM month, is outside). The downcurve regime is a different matter, since it was located by asking where the signal's false alarms sat, and the paper treats its two-era evidence accordingly.

Table E.1: The headline in the two halves of the sample. State regressions $y_t = a + b\,\mathbf{1}\{\text{signal}\}_{t-1} + e_t$ (%/yr; no controls) estimated separately on 1988:01–2004:12 and 2005:01–2026:02, with the episode-clustered t and the wild cluster bootstrap p on the half's signal episodes (B = 9,999); stars by the wild p. The footer reports each half's months, signal months, episodes, and how many of its three worst carry months fall inside the lagged state.

	1988:01–2004:12			2005:01–2026:02		
	b	t_{ep}	wild p	b	t_{ep}	wild p
Carry, spot	-11.41^{**}	-2.54	0.037	-15.85^{**}	-1.99	0.010
Carry, total	-10.20^{*}	-2.04	0.052	-14.76^{**}	-1.84	0.047
High-beta, spot	-4.01	-1.12	0.287	-11.91^{***}	-4.07	0.010
High-beta, total	-3.03	-0.80	0.448	-11.54^{**}	-3.85	0.011
Months	204			254		
Signal months	54			38		
Episodes	7			8		
Three worst carry months inside	2 of 3			3 of 3		

The common term premium. The expectations reading of Section 2.2 has one well-known rival, a common global term premium that could invert curves together without any easing being expected. The check uses the U.S. ACM slope term premium (the ten-year less the two-year premium of Adrian et al., 2013), the standard proxy for the common component, in two ways. First, as a control. Adding its lagged level, or its level and one-month change, to the headline regression leaves every cell where it was (carry spot -12.9%/yr, wild $p = 0.001$; high-beta spot -7.6, $p = 0.011$). Second, by stripping it from the curves. Each country's slope is regressed on the U.S. premium and its fitted exposure removed; the loadings are small (between -0.38 for Switzerland and $+0.23$ for Canada, and within 0.05 of zero for Australia, New Zealand, the euro, and Japan), 83% of the 1980–2026 slope panel's inversion months remain inverted, and the IYC rule rebuilt on the stripped slopes produces a state that contains all five worst carry months, keeps the high-beta loss (-6.2%/yr in spot, wild $p = 0.004$), and keeps the downcurve split, with the loss in its non-downcurve months and a positive wedge. The rebuilt state is broader than the baseline (125 months against 92, sharing 83), and on that broader state the carry portfolio's loss is about half the baseline's and not significant (-5.4%/yr, $p = 0.11$). Appendix K reports the exhibit. A common premium of this kind therefore does not manufacture the inversions or the high-beta loss;

what the proxy cannot settle is the country-specific premium, which no comparable series in the paper measures.

The downcurve regime. The wedge's definitional sensitivities are disclosed alongside the result (Section 5.3). The wedge is robust to how the regime is drawn at tight definitions, priced by the shape rather than the rate level, and conditional on the signal rather than an unconditional return predictor. It is worth restating here that the regime's second era (2023–24) sits mostly outside the signal state and still shows the compensation pattern (+3.5%/yr in non-signal downcurve months)—the closest available out-of-sample check.

Timing. Every regressor in the paper sits in the $t-1$ information set, and the claim is executed rather than asserted. A truncation audit rebuilds the signal from data ending at each of 435 month-ends and finds that no month's classification ever changes when later data arrives; the same holds for the inversion counts and all nine own signals. A second audit recomputes every conditioning series from truncated data—the downcurve regime, the naive count and its age, the U.S. inversion, the global slope and its change, the controls, the Fama differential, the carry sort, and the expanding equity betas of both the G10 and the sixteen-currency universes—and finds every value dated T identical to its full-sample counterpart, with every portfolio's month-(T+1) legs fixed by end-of-T data (492 membership checks, zero mismatches). Real-time availability is equally clean. Every conditioning variable is market-priced at month-end except the NFCI, a weekly series released within a week; CPI appears only in descriptive tables, at its two-month release lag, and the activity series of Section 4 are outcomes, not conditioning. Lagging the signal one *additional* month halves the carry coefficient, shrinks the high-beta one, and costs significance (carry spot -12.9 to -6.3; high-beta -7.5 to -4.7)—the forward-decay profile an ex-ante state variable signal should show—and forward-shift placebos destroy the results entirely. The 2-year/10-year construction keeps the expected-path component of the slope dominant (Appendix C shows the 3-month tenor is not a substitute), and the paper accordingly leads with risk measurement rather than tradable alpha.

F Robustness: Adding the U.S. Curve and the Dollar

The paper's constructions deliberately exclude the United States. The U.S. curve does not enter the count (Section 6 shows it carries no incremental information), and the dollar is the numeraire rather than a portfolio currency. Table F.1 repeats the headline cells with the U.S. included on both margins, through a *ten-curve signal* counting the U.S. 2s10s alongside the nine (identical entry and exit rules) and a *dollar-inclusive portfolio* of ten currencies sorted on rate levels, in which the dollar contributes zero spot and zero income when held.

Table F.1: Headline cells with the U.S. included in the signal count and/or the portfolio. State regressions $y_t = a + b\,\mathbf{1}\{\text{signal}\}_{t-1} + e_t$; wild bootstrap p on signal runs, B = 9,999; no controls (the controlled versions are in the replication package).

Portfolio	Signal	Leg	b	t_{run}	wild p
G10 carry	Baseline (9 curves)	spot	-12.85^{***}	-3.03	0.001
G10 carry	Baseline (9 curves)	total	-11.31^{***}	-2.40	0.003
G10 carry	+US curve (10)	spot	-11.72^{***}	-2.92	0.001
G10 carry	+US curve (10)	total	-10.20^{***}	-2.34	0.006
Dollar-incl.	Baseline (9 curves)	spot	-13.11^{***}	-2.69	$<$0.001
Dollar-incl.	Baseline (9 curves)	total	-11.54^{***}	-2.15	0.009
Dollar-incl.	+US curve (10)	spot	-12.10^{***}	-2.68	0.001
Dollar-incl.	+US curve (10)	total	-10.57^{**}	-2.14	0.013

Signal months: baseline (9 curves) 92, +US curve (10) 101, overlap 92; USD sits in the funding leg 31% of months

Nothing changes. The baseline signal's 92 months are a strict subset of the ten-curve signal's 101—adding the U.S. curve adds nine months and subtracts none—and every combination of portfolio and signal remains significant (bare wild $p < 0.001$–0.013; controlled 0.002–0.055). The dollar-inclusive portfolio is marginally *more* exposed on the spot leg, with the dollar sitting in the funding leg roughly a third of the time. Excluding the U.S. from the main text is a presentational choice, not a load-bearing one.

G Beta-Definition Robustness

The high-beta portfolio sorts on the currency beta of (3), estimated monthly on an expanding window against the S&P 500. Table G.1 rebuilds the portfolio with the same expanding estimator against the MSCI World index instead. On the post-1999 subsample the two definitions agree, since every filtered cell is negative and significant at the five percent level under either index. On the full sample the MSCI definition loses significance (p from 0.16 to 0.54), and the reason is the index's own composition. Around 1990 Japan was roughly two-fifths of MSCI World, so the yen's beta against that index is dominated by its home market's weight, and on 1989–99 the MSCI definition ranks the yen and the franc at the top of the beta ordering, inverting the growth-exposure ranking and putting the funding currencies in the long leg. The baseline uses the S&P 500 precisely because an external market index measures a currency's exposure to world risk without home-index contamination. Once Japan's weight falls below a fifth, by 1999, the two definitions agree.

Table G.1: The high-beta portfolio under alternative currency-beta definitions. The expanding monthly estimator of (3) against the S&P 500 (baseline) and against the MSCI World index; sorts lagged one month. Cells are state regressions $y_t = a + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_t$ of the resulting portfolio's spot and total returns on the signal and on the filtered signal, with the wild bootstrap p on signal episodes. Panel A uses each definition's maximal all-nine-beta sample (the baseline row therefore differs slightly from Table 4, which also uses the fifteen early months with seven or eight available betas); Panel B the post-1999 subsample, on which Japan's weight in MSCI World has fallen below one-fifth.

		Signal				Signal excl. downcurve			
		Spot		Total		Spot		Total	
Beta definition	N	b	wild p	b	wild p	b	wild p	b	wild p
Panel A: each definition's maximal sample									
Monthly expanding, S&P 500 (baseline)	443	−7.93***	0.005	−7.19***	0.009	−12.31***	0.001	−11.97***	0.001
Monthly expanding, MSCI World	443	−2.93	0.319	−4.00	0.162	−2.30	0.544	−2.65	0.446
Panel B: post-1999 subsample (Japan below one-fifth of MSCI World)									
Monthly expanding, S&P 500 (baseline)	320	−11.10***	0.003	−10.73***	0.004	−13.35***	0.001	−13.09***	0.001
Monthly expanding, MSCI World	320	−7.67**	0.030	−7.85**	0.019	−8.29**	0.014	−8.67***	0.008

H Portfolio Composition by State

Table H.1 reports, for each currency, the share of months it occupies each leg of the carry and high-beta portfolios—in the full sample, in the IYC-cluster months, and in the cluster's non-downcurve and downcurve months. The feature that matters is invariance. The core of each leg barely moves across states (the minor slots rotate—the krona, euro, and franc trade places in the high-beta short leg across the columns—but the anchors do not). The Australian and New Zealand dollars anchor the carry long leg at nearly identical shares in and out of the cluster (91/73% against 85/71%), the franc and the yen hold the short leg in effectively every cluster month, and the Canadian dollar sits in the high-beta long leg in every month of the sample. The conditional results in the paper therefore reprice a *standing* portfolio; they are not generated by the sort rotating into different currencies when the signal fires. The one visible shift is the third long-leg slot in the downcurve column—the pound (97%) displaces the krone—and it is era composition, because the 35 downcurve cluster months sit in 1988–91 (29 of the 35), 2007:09, and 2023, when U.K. rates were near the top of the ranking, and the Deutsche-mark-era euro correspondingly fills the short leg there. The top-two anchor of each leg is unchanged across the split, so the downcurve wedge of Section 5.3 is priced on the same core portfolio.

I The EM-Native Regime Check

Section 7 reports that the downcurve regime travels to emerging markets in shape but not in pay. Table I.1 contains the underlying cells, in which everything is constructed EM-natively—the signal is an inversion count over the seven EM curves (≥ 2 of those available, at least four required), the regime is the EM average curve's own shape with the money-market rate as the short point, and the returns are the equal-weighted EM basket's (2003:07–2026, when all seven EM curves are live). The EM downcurve months coincide with the G10's within the overlapping sample (mid-2007 and 2023); the EM own-signal costs the basket

Table H.1: Portfolio composition by state, measured as the share of months (%) each currency occupies the long (top-3) and short (bottom-3) leg of the G10 carry portfolio (Panel A) and the high-beta portfolio (Panel B), in the full sample, the IYC-cluster months, and the cluster's non-downcurve (Excl. DC) and downcurve (DC) months; 1988:01–2026:02, all states lagged. Sorting variables are as in the text, the lagged money-market rate and the expanding equity beta (3).

	Long leg (% of months)				Short leg (% of months)			
	Full	IYC cluster	Excl. DC	DC	Full	IYC cluster	Excl. DC	DC
Panel A: the carry portfolio (rate sort)								
NZ	85	91	93	89	2	1	2	0
AU	71	73	72	74	4	0	0	0
NO	62	34	53	3	3	5	9	0
GB	43	70	53	97	4	0	0	0
CA	20	21	25	14	23	14	21	3
SE	15	12	5	23	33	36	49	14
EU	4	0	0	0	51	47	25	83
JP	0	0	0	0	83	97	95	100
CH	0	0	0	0	96	100	100	100
Panel B: the high-beta portfolio (equity-beta sort)								
CA	100	100	100	100	0	0	0	0
AU	97	92	96	86	0	0	0	0
NZ	77	76	74	80	0	0	0	0
NO	26	32	30	34	0	0	0	0
EU	0	0	0	0	46	45	61	17
GB	0	0	0	0	52	47	46	49
SE	0	0	0	0	34	59	44	83
JP	0	0	0	0	73	65	53	86
CH	0	0	0	0	95	85	96	66

Months per state: full 458,
IYC cluster 92, excl. downcurve 57, downcurve 35.

−10.6%/yr in non-downcurve months; and in the downcurve months the basket is roughly flat—but the EM interest rate differential there (+3.4%/yr) sits *below* its own norm rather than above it, so the regime defangs the signal without paying the position.

J Local Projections and the Persistence of the State

The signal state is a run of consecutive months (fifteen runs, ninety-two months), so a projection of the h-month change on the lagged state averages the path after every signal month, including months deep in a run whose window overlaps with the run's own continuation. The wild cluster bootstrap on runs handles the dependence for inference but not the inter-

Table I.1: EM-native construction. The EM basket's returns (annualized, %/yr) by the EM own-signal state and the EM average curve's shape, 2003:07–2026. Wedge regression $y_t = a + b_1\,\mathbf{1}\{\text{signal}\}_{t-1} + b_2\,\mathbf{1}\{\text{signal}\}_{t-1} \times \mathbf{1}\{\text{downcurve}\}_{t-1} + e_t$ on EM-signal episodes; t_{NW} = Newey–West (12 lags), t_{ep} = episode-clustered. The seven-month downward-sloping cell is suggestive, not conclusive.

Cell	N	Spot	Income	Total
EM signal on, EM curve downward-sloping	7	−2.01	+3.39	+1.38
EM signal on, other shapes	71	−10.20	+4.68	−5.52
EM signal off	194	+0.60	+4.47	+5.07

Average EM curve in downcurve months: money market 8.38, 2Y 7.94, 10Y 7.81 (%)
Wedge regression (total): EM signal in other shapes $b = -10.59$ ($t_{NW} = -2.34$, $t_{ep} = -2.90$); signal × downward-sloping $b = +6.91$ ($t_{NW} = +1.45$, $t_{ep} = +2.14$)

pretation. Figure J.1 therefore repeats the projections of Figure 6 under two designs that do not average over the continuation, the onset design, in which the regressor is one only in the first month of each run, and the controlled design, in which the state enters together with its own lag and three lags of the outcome's monthly change, so the coefficient is the response to the state's arrival given the previous month's state and the outcome's recent momentum. The leading indicator's decline is the same under both designs at six months and larger at its seventh-month trough, the spot leg's loss is larger under both, the money-market rate is flat under all three, and industrial production is the one series whose decline does not survive the controls (the estimates are reported in Section 4).

K The Common Term Premium

Appendix E confronts the expectations reading of the inversions with a common global term premium, proxied by the U.S. ACM slope premium of Adrian et al. (2013), the ten-year less the two-year premium, the standard proxy for the common component. Table K.1 reports the exhibit. Each country's 10Y−2Y slope is regressed on the U.S. premium over the slope panel (1980:01–2026:02) and its fitted exposure is removed, the IYC rule of Section 2.2 is rebuilt on the stripped slopes with the same entry, exit, and threshold, and the rebuilt state

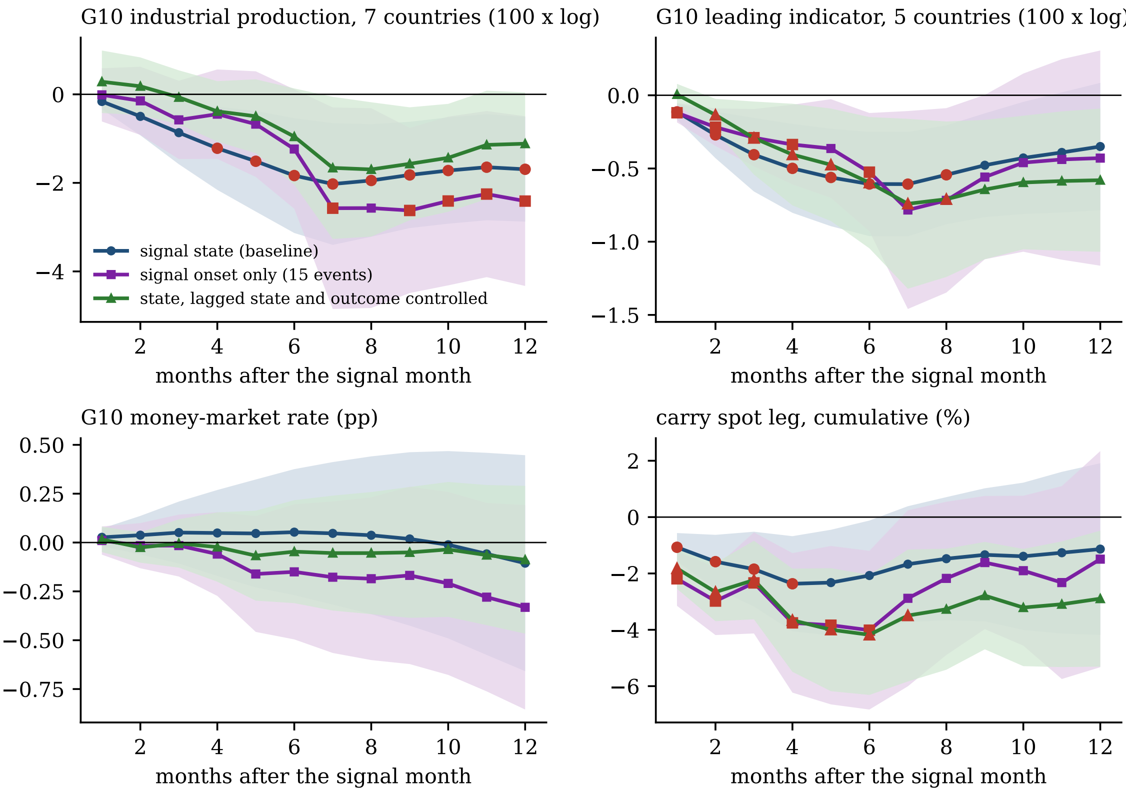

Figure J.1: Three designs for the local projections of Figure 6. Circles are the baseline state design, squares the onset design (the first month of each signal run, fifteen events), and triangles the controlled design (the state with its own lag and three lags of the outcome's monthly change); 1988:01–2026:02. Bands are 90% Newey–West intervals with $h + 1$ lags, and red markers have a wild cluster bootstrap p below 0.10 with signal runs as clusters.

is lagged one month and split by the downcurve regime of (17). The loadings are small, between −0.38 (Switzerland) and +0.23 (Canada) and within 0.05 of zero for Australia, New Zealand, the euro, and Japan, so the strip removes little. Of the 1,134 curve-months of inversion in the slope panel (1980:01–2026:02), 942 (83%) remain inverted. The rebuilt state has 125 months in 18 runs, against the baseline's 92 in 15, and shares 83 of them; it contains all five worst carry months (2008:10, 1995:03, 2020:03, 2008:09, and 1998:10), as the baseline does.

The table's rows are the state regressions of Table 4, each cell against all other months, with the within-signal downcurve wedge estimated on the rebuilt state's runs. The high-beta portfolio loses on the rebuilt state as it does on the baseline (−6.2%/yr in spot, wild

Table K.1: The signal rebuilt on term-premium-stripped slopes. Each country's 10Y−2Y slope is stripped of its fitted exposure to the U.S. ACM slope term premium, the IYC rule is rebuilt on the stripped slopes and lagged one month, and the rebuilt state is split by the downcurve regime. Cells are state regressions $y_t = a + b\,\mathbf{1}\{\text{state}\}_{t-1} + e_t$ of the carry and high-beta portfolios' spot and total returns (annualized, %/yr) against all other months; the wedge row is the within-signal regression of the return on the downcurve dummy on the rebuilt state's months. Wild cluster bootstrap p on the state's runs in brackets ($B = 9{,}999$), stars by that p; 1988:01–2026:02.

State	Carry, spot	Carry, total	High-beta, spot	High-beta, total
Baseline signal	-12.85^{***}	-11.31^{***}	-7.53^{***}	-6.82^{**}
	[0.001]	[0.003]	[0.010]	[0.013]
Stripped signal	-5.42	-4.07	-6.17^{***}	-5.66^{***}
	[0.107]	[0.251]	[0.004]	[0.008]
non-downcurve months	-7.68^{*}	-7.03	-10.27^{***}	-10.28^{***}
	[0.094]	[0.142]	[0.000]	[0.000]
downcurve months	$+1.56$	$+3.81^{*}$	$+5.00^{**}$	$+6.36^{**}$
	[0.621]	[0.059]	[0.026]	[0.012]
downcurve wedge	$+7.64$	$+9.18^{*}$	$+12.89^{***}$	$+14.16^{***}$
	[0.149]	[0.094]	[0.007]	[0.004]

$p = 0.004$; -5.7 in total, $p = 0.008$), and the carry portfolio's loss is about half its baseline value, -5.4%/yr in spot at $p = 0.11$, on the broader state. The downcurve split survives. The rebuilt state's non-downcurve months carry the loss (carry spot -7.7, $p = 0.09$; high-beta spot -10.3, $p < 0.001$), its downcurve months are flat to positive (carry spot $+1.6$; high-beta spot $+5.0$, $p = 0.03$), and the wedge is positive in every column, significant for the high-beta portfolio (spot $+12.9$, $p = 0.007$) and at the ten-percent level for carry total returns ($+9.2$, $p = 0.09$). The proxy is a U.S. series, so the test addresses a common premium and not country-specific premia, which no comparable series in the paper measures.